

Welcome (Again!) to
MATH 4100/COMP 5360 – Introduction to Data Science

Data Science Ethics

February 19, 2026

While we are waiting for class to start, think
about your answer to:

What are ethics?

What are ethics?

What are ethics?

- Shared principles and values that help us distinguish right from wrong.
- Cornerstone of civilization
- The basis for the rules we all voluntarily choose to follow for the betterment of all of us.

Example: We don't steal due to our ethics.

- Even if we can get away with it!



Ethics are not Laws

Is it **legal** for me to lie to you?

Is it **ethical**?

Ethics are not Laws

Is it **legal** for me to lie to you?

Is it **ethical**?

- Just because something is legal, does not mean it is ethical.
- Just because something is ethical, does not mean it is legal.

**Ethics are something you have to
decide for yourself.**

**Ethics are something you have to
decide for yourself.**

**...but often policies follow agreed
upon ethics.**

Abuses of human subjects in biomedical research led to modern codes of research ethics

Nazi experimentation on WWII concentration camp prisoners.

- Nuremberg Code
- Declaration of Helsinki



[Image from](http://broughttolife.sciencemuseum.org.uk/broughttolife/techniques/nurembergcode)

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Abuses of human subjects in biomedical research led to modern codes of research ethics

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USA, Tuskegee Study of Untreated Syphilis in Black Men

- National Commission for the Protection of Human Subjects of Biomedical & Behavior Research
- Belmont report

When did these experiments come to an end?



**Nuremburg hearings on Nazi
concentration camp experiments**



**Tuskegee Study of Untreated
Syphilis in Black Men**

Image from
<http://broughttolife.sciencemuseum.org.uk/broughttolife/techniques/nurembergcode>

https://en.wikipedia.org/wiki/Tuskegee_syphilis_experiment#/media/File:Tuskegee-syphilis-study_doctor-injecting-subject.jpg

Abuses of human subjects in biomedical research led to modern codes of research ethics

Nazi experimentation on WWII concentration camp prisoners.

- Nuremberg Code (1947)
- Declaration of Helsinki (1964)

USA, Tuskegee Study of Untreated Syphilis in Black Men

- National Commission for the Protection of Human Subjects of Biomedical & Behavior Research (1974)
- Belmont report (1979)

“But we’re data scientists.”

“We’re not doing medical experiments”

“But we’re data scientists.”

“We’re not doing medical experiments”

There are risks beyond medical experiments.

Social science research also shown to carry risks of harm

- Milgram’s Obedience to Authority study



Adapted from CITI Program, Image from <http://content.time.com/time/health/article/0,8599,1867735,00.html>

Public Announcement

**WE WILL PAY YOU \$4.00 FOR
ONE HOUR OF YOUR TIME**

Persons Needed for a Study of Memory

*We will pay five hundred New Haven men to help us complete a scientific study of memory and learning. The study is being done at Yale University.
*Each person who participates will be paid \$4.00 (plus 50c carfare) for approximately 1 hour's time. We need you for only one hour: there are no further obligations. You may choose the time you would like to come (evenings, weekdays, or weekends).

*No special training, education, or experience is needed. We want:

Factory workers	Businessmen	Construction workers
City employees	Clerks	Salespeople
Laborers	Professional people	White-collar workers
Barbers	Telephone workers	Others

All persons must be between the ages of 20 and 50. High school and college students cannot be used.
*If you meet these qualifications, fill out the coupon below and mail it now to Professor Stanley Milgram, Department of Psychology, Yale University, New Haven. You will be notified later of the specific time and place of the study. We reserve the right to decline any application.
*You will be paid \$4.00 (plus 50c carfare) as soon as you arrive at the laboratory.

TO:
PROF. STANLEY MILGRAM, DEPARTMENT OF PSYCHOLOGY,
YALE UNIVERSITY, NEW HAVEN, CONN. I want to take part in
this study of memory and learning. I am between the ages of 20 and
50. I will be paid \$4.00 (plus 50c carfare) if I participate.

NAME (Please Print)

ADDRESS

TELEPHONE NO. Best time to call you

AGE OCCUPATION SEX

CAN YOU COME:

WEEKDAYS EVENINGS WEEKENDS

Social science research also shown to carry risks of harm

- Milgram's Obedience to Authority study (1961)
- Zimbardo's Stanford Prison experiment (1971)



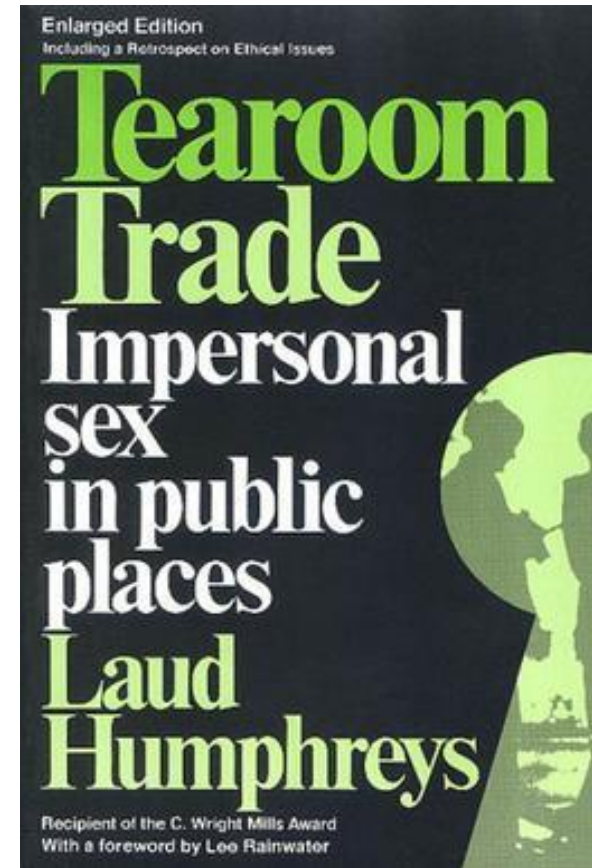
Adapted from CITI Program, image from
<https://www.insidehighered.com/news/2018/06/20/new-stanford-prison-experiment-revelations-question-findings>

Social science research also shown to carry risks of harm

- Milgram's Obedience to Authority study (1961)
- Zimbardo's Stanford Prison experiment (1971)
- Humphreys' Tearoom Trade study (1970)

Risks:

- Psychological well-being
- Privacy
- Autonomy
- Reputation



“Tastes, Ties, and Time” Study



Collected data about students, matching Facebook data and university records.

Dataset was released publicly with names removed.

➤ However, it was not difficult to de-anonymize.

Students were not informed of this study.

Would you be comfortable with this?

What if it was your internet search history?

Adapted from CITI Program, “Five Ethical Provocations in HCI Research” (Brown et al. 2016), &
<https://www.chronicle.com/article/Harvards-Privacy-Meltdown/128166>

“Tastes, Ties, and Time” Study



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Collected data about students, matching Facebook data and university records.

➤ Student workers helped collect the Facebook data, some of whom may have privileged access to Harvard-only groups and settings

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Who owns data? Who controls data?

theverge.com/2023/1/17/23558516/ai-art-copyright-stable-diffusion-getty-images-lawsuit

Getty Images is suing the creators of AI art tool Stable Diffusion for scraping its content / Getty Images claims Stability AI 'unlawfully' scraped millions of images from its site. It's a significant escalation in the developing legal battles between generative AI firms and content creators.



https://www.npr.org/2025/01/14/nx-s1-5258952/new-york-times-openai-m

BUSINESS

'The New York Times' takes OpenAI to court. ChatGPT's future could be on the line

UPDATED JANUARY 14, 2025 · 4:27 PM ET ⓘ



Bobby Allyn



JAMES VINCENT

17, 2023, 3:30 AM MST | □



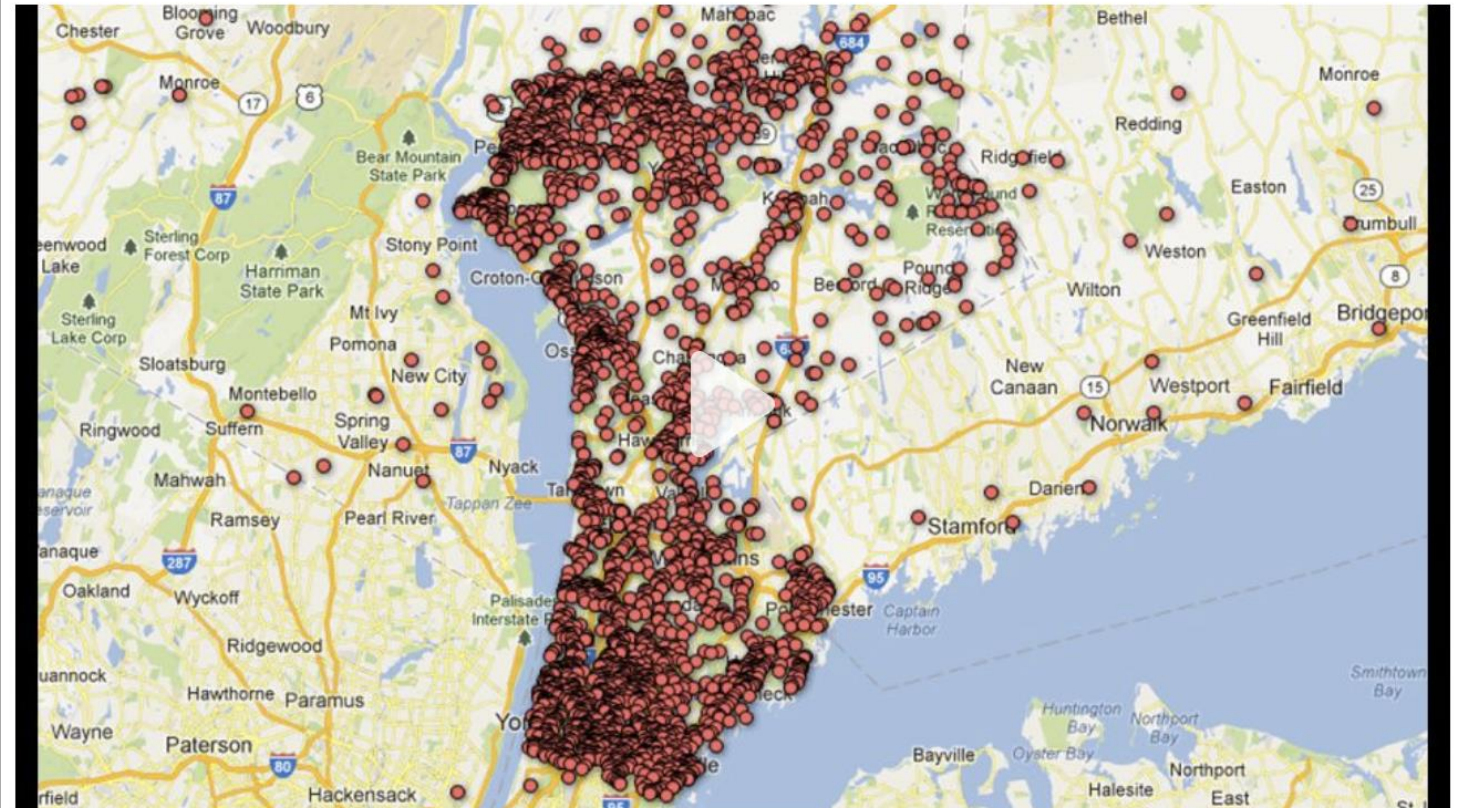
Who owns data? Who controls data?

What's private?

Newspaper sparks outrage for publishing names, addresses of gun permit holders

KC Maas and Josh Levs, CNN

Updated 10:23 AM EST, Thu December 27, 2012



Video Ad Feedback

02:40 - Source: [CNN](#)

Newspaper's gun permit map causes outcry

Is Facebook purposefully making you sad?

Facebook experiment biased shown posts to be either more positive or more negative to see how users would react.

TECHNOLOGY

Facebook Tinkers With Users' Emotions in News Feed Experiment, Stirring Outcry

By VINDU GOEL JUNE 29, 2014



216



Facebook revealed that it had altered the news feeds of over half a million users in its study. Karen Bleier/Agence France-Presse — Getty Images

To [Facebook](#), we are all lab rats.

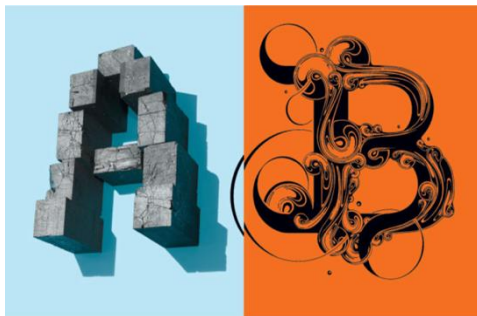
Facebook routinely adjusts its users' news feeds — testing out the number of ads they see or the size of photos that appear — often without their knowledge. It is all for the purpose, the company says, of creating a more alluring and useful product.

But last week, Facebook revealed that it had manipulated the news feeds of over half a million randomly selected users to change the number of positive and negative posts they saw. It was part of a psychological study to

Is Facebook purposefully making you sad?

Facebook experiment biased shown posts to be either more positive or more negative to see how users would react.

Is this “just an A/B test”?



SECTIONS

The New York Times

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Ethical Principles in Human Subjects Research

The Belmont Report put forth three key principles:

- Respect for persons
- Beneficence
- Justice





Respect for Persons

“individuals should be treated as autonomous agents, and second, that persons with diminished autonomy are entitled to protection”

- **Autonomy** means people must be able to make informed decisions about their actions and well-being.
- Participation in research should be **Voluntary at all times**.
- **Informed Consent** is the process by which people exercise their autonomy to volunteer to participate.
- Included in the informed consent should be notification of any risks or harm that could occur.

Beneficence

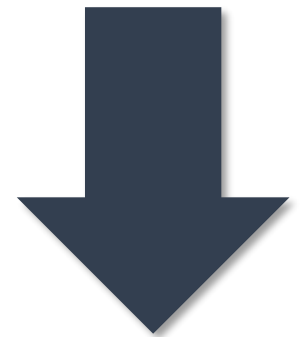
Beneficence is an obligation to not only protect participants from harm, but “maximize possible benefits and minimize possible harms.”

Common risks in non-medical research can include:

- Psychological distress
- Fatigue
- Loss of privacy
- Reputational harm



Benefits



Harms



Justice

Justice is an obligation that those who bear the burdens of the research should not differ from those who receive the benefits.

Historically, often the poor participated in the experiments while the rich received the rewards.

Similarly, prisoners have been used as subjects with no hope of receiving the rewards.

Research experiments with human subjects require ethical review

Research organizations, like the University of Utah, have **Institutional Review Boards (IRB)** which must approve of proposed research before it can go forward.



The IRB considers the **balance** of respect for persons, beneficence, and justice when making their determination.

- Review includes informed consent documents
- Harms and benefits included in application
- Handling of personally identifiable information (privacy) also considered.

IRB approval **does not mean an experiment is ethical.**

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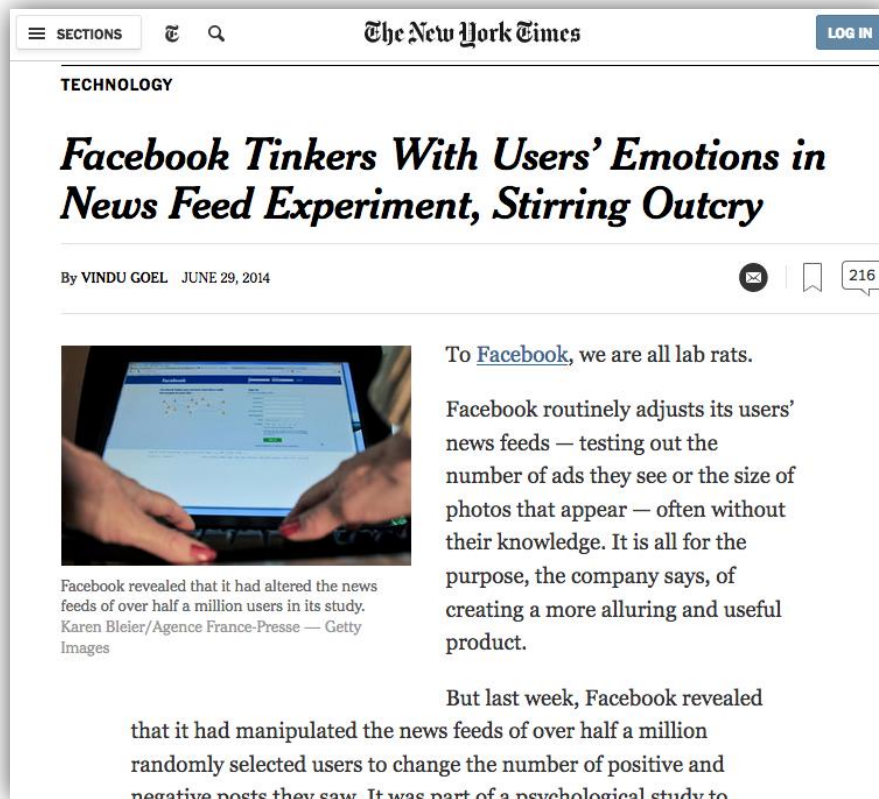
Lack of IRB approval **does not mean an experiment is unethical.**

IRB approval **does not mean an experiment is ethical.**

Lack of IRB approval **does not mean an experiment is unethical.**

IRB approval indicates to the scientific community that a panel of experts found the experiment to be consistent with ethical standards.

IRB approval **does not** mean an experiment is ethical.



This allegedly had IRB approval at one point...

But...

Is Facebook trying to make you sad?

IRB only consulted after experiment was run.
Unclear if consent was in terms of service.

➤ Is consent in terms of service informed consent?

SECTIONSE

THE NEW YORK TIMES

LOG IN

TECHNOLOGY

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Some initial concerns...

The following issues are frequent:

- Privacy, particularly in light of personalization & social media.
- Informed consent and volunteering
- Psychological well-being
 - With personal data or in special settings (e.g., medical apps & devices)
 - Frustration, Fatigue, Discovering you are color blind
- Applicability across cultural and demographic groups

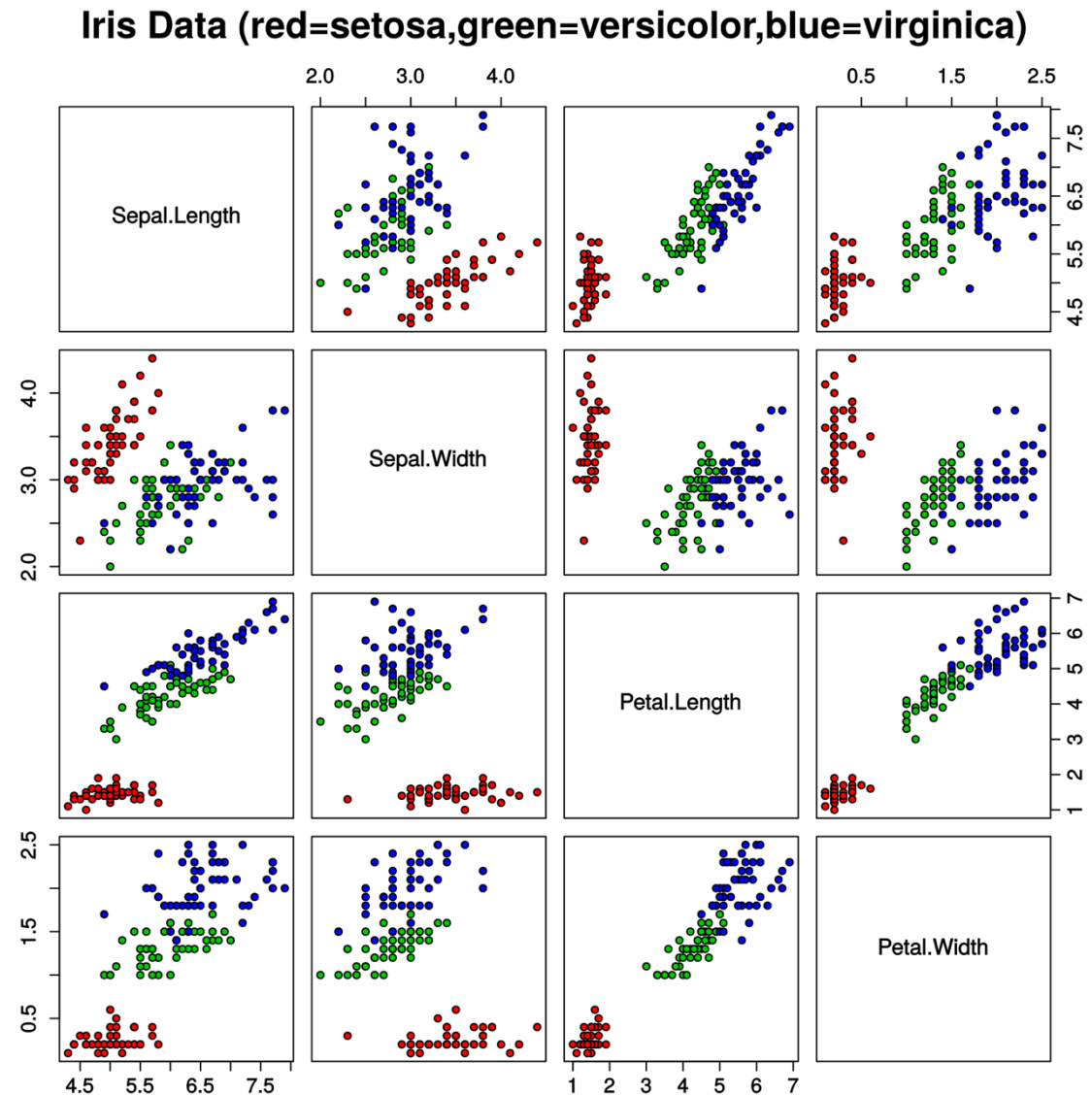


See Ethical Encounters in HCI Workshops:
<https://ethicalencountershci.com>

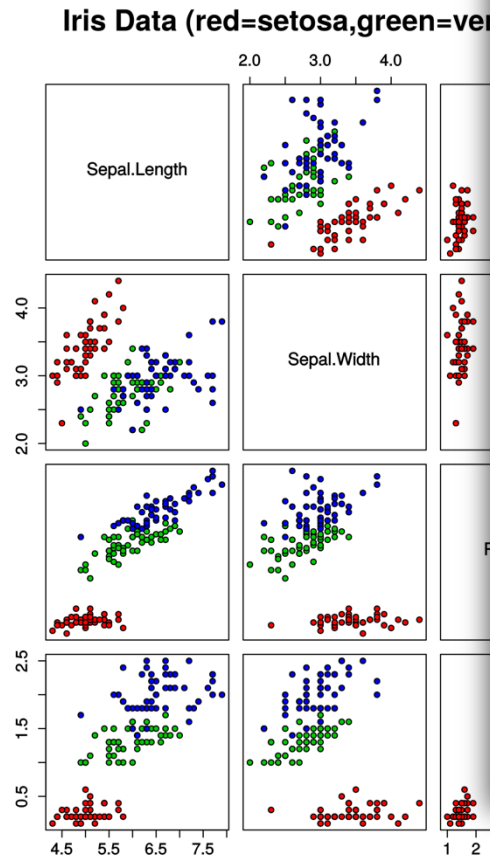
Image from <https://www.agilitypr.com/pr-news/public-relations/digital-behavior-powers-culture-across-demographic-segments/>

Data Science and Societal Consequence

Show of hands: How many of you are familiar with the Iris dataset?



Do you know where Fisher's iris data set was initially published?



Annals of Eugenics / Volume 7, Issue 2 / p. 179-188

[Free Access](#)

THE USE OF MULTIPLE MEASUREMENTS IN TAXONOMIC PROBLEMS

R. A. FISHER Sc.D., F.R.S.

First published: September 1936

<https://doi.org/10.1111/j.1469-1809.1936.tb02137.x>

Citations: 8,332

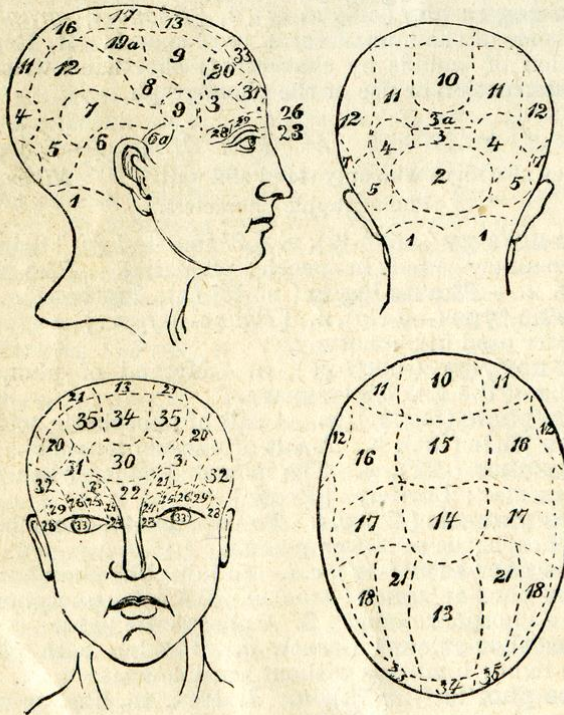
[About](#)

Abstract

The articles published by the Annals of Eugenics (1925–1954) have been made available online as an historical archive intended for scholarly use. The work of eugenicists was often pervaded by prejudice against racial, ethnic and disabled groups. The online publication of this material for scholarly research purposes is not an endorsement of those views nor a promotion of eugenics in any way.

Phrenology, 1895

Phre-nol'o-gy (-nŏl'ŏ-jŏ), *n.* [Gr. φρήν, φρενός + -logy.] **1.** Science of the special functions of the several parts of the brain, or of the supposed connection between the faculties of the mind and organs in the brain. **2.** Physiological hypothesis that mental faculties, and traits of character, are shown on the surface of the head or skull; craniology. — **Phre-nol'o-gist**, *n.* — **Phren'o-log'ic** (frĕn'ŏ-lŏj'ĭk), **Phren'o-log'ic-al**, *a.*

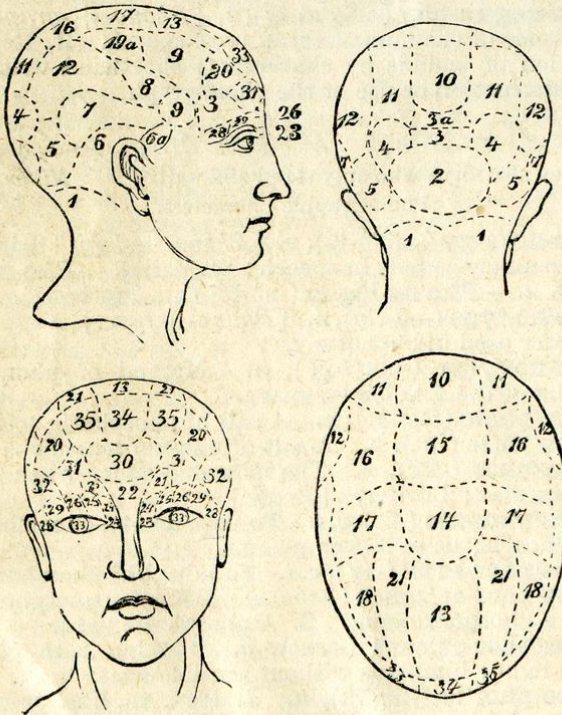


A Chart of Phrenology.

1 Amativeness ; **2** Philoprogenitiveness ; **3** Concentrativeness ; **3 a** Inhabitativeness ; **4** Adhesiveness ; **5** Combaticiveness ; **6** Destructiveness ; **6 a** Alimentiveness ; **7** Secretiveness ; **8** Acquisitiveness ; **9** Constructiveness ; **10** Self-esteem ; **11** Love of Approbation ; **12** Cautiousness ; **13** Benevolence ; **14** Veneration ; **15** Firmness ; **16** Conscientiousness ; **17** Hope ; **18** Wonder ; **19** Ideality ; **19 a** (Not determined) ; **20** Wit ; **21** Imitation ; **22** Individuality ; **23** Form ; **24** Size ; **25** Weight ; **26** Coloring ; **27** Locality ; **28** Number ; **29** Order ; **30** Eventuality ; **31** Time ; **32** Tune ; **33** Language ; **34** Comparison ; **35** Causality. [Some raise the number of organs to forty-three.]

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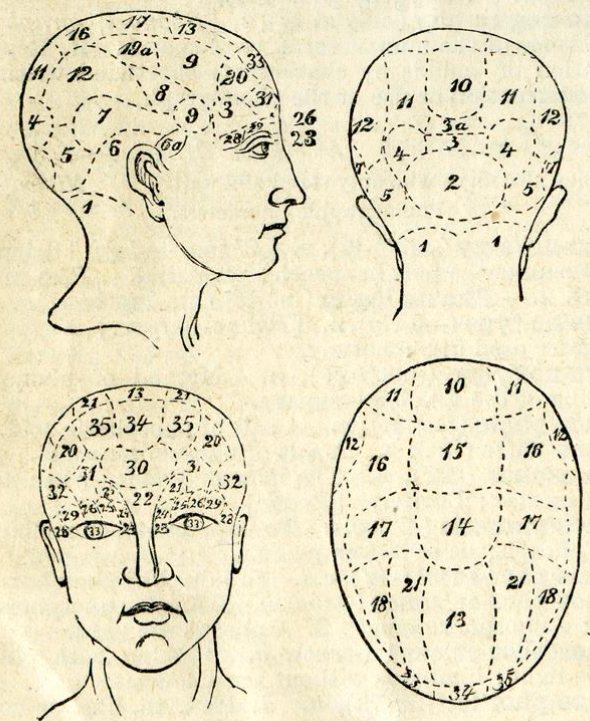
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Baked into this practice is an **assumption** that it could work.

It's all too easy to torture the data to say what one wants it to say.

Phrenology, 2016

Phre-nol'o-gy (-nōl'ō-jy), *n.* [Gr. φρήν, φρενός + -logy.] **1.** Science of the special functions of the several parts of the brain, or of the supposed connection between the faculties of the mind and organs in the brain. **2.** Physiological hypothesis that mental faculties, and traits of character, are shown on the surface of the head or skull; craniology. — **Phre-nol'o-gist**, *n.* — **Phren-o-log'ic** (frēn'ō-lōj'ik), **Phren'o-log'ic-al**, *a.*



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arXiv:1611.04135v2 [cs.CV] 21 Nov 2016

Automated Inference on Criminality using Face Images

Xiaolin Wu
McMaster University
Shanghai Jiao Tong University
xwu510@gmail.com

Xi Zhang
Shanghai Jiao Tong University
zhangxi_19930818@sjtu.edu.cn

Abstract

We study, for the first time, automated inference on criminality based solely on still face images, which are free from any biases of subjective judgments of human observers. Via supervised machine learning, we build four classifiers (logistic regression, KNN, SVM, CNN) using face images of 1856 real persons controlled for race, gender, age and facial expressions, nearly half of whom were criminals, for discriminating between criminals and non-criminals. All four classifiers perform consistently well to empirically establish the validity of automated face image inference on criminality, despite the historical controversy surrounding this line of enquiry. Also, some distinctive structural features for predicting criminality are discovered by machine learning. Above all, the most important discovery of this research is that criminal and non-criminal face images populate two quite distinctive manifolds. The variation among criminal faces is significantly greater than that of the non-criminal faces. The two manifolds of criminal and non-criminal faces appear to be well separated, with the non-criminal manifold lying in a lower-dimensional space with a smaller span, exhibiting a law of "normality" for faces of non-criminals. In other words, the faces of the general law-biding public have a greater degree of resemblance compared with the faces of criminals, or criminals have a higher degree of dissimilarity in facial appearance than non-criminals.



(a) Three samples in criminal ID photo set S_c .



(b) Three samples in non-criminal ID photo set S_n .

Figure 1. Sample ID photos in our data set.

of static face images [35]. This method can synthesize a representative (average) face image for a set of input face images scored closely on any of the four aforementioned social percepts. The ranking of these synthesized face im-

Data Science allows us to discriminate so much faster now.

Is it okay for employers to do an internet search on interviewees?

Google Targeted Advertising – Name Search

Ad related to latanya sweeney ⓘ

Latanya Sweeney Truth

www.instantcheckmate.com/

Looking for **Latanya Sweeney**? Check **Latanya Sweeney's** Arrests.

Ads by Google

Latanya Sweeney, Arrested?

1) Enter Name and State. 2) Access Full Background Checks Instantly.

www.instantcheckmate.com/

Latanya Sweeney

Public Records Found For: **Latanya Sweeney**. View Now.

www.publicrecords.com/

La Tanya

Search for La Tanya Look Up Fast Results now!

www.ask.com/La+Tanya

Ads by Google

Kirsten Lindquist

Get **Kirsten Lindquist** Find **Kirsten Lindquist**

www.ask.com/Kirsten+Lindquist

We Found:Kristen Lindquist

1) Contact **Kristen Lindquist** - Free Info! 2) Current Phone, Address & More.

www.peoplesmart.com/

Search by Phone

Background Checks

Public Records

Search by Email

Search by Address

Criminal Records

Kristen Lindquist

Public Records Found For: **Kristen Lindquist**. View Now.

www.publicrecords.com/

instantcheckmate.com advertised for Latanya, not Kirsten, but...



[DASHBOARD](#) | [EDIT ACCOUNT INFO](#) | [LOGOUT](#)



LATANYA SWEENEY

1420 Centre Ave
Pittsburgh, PA 15219
DOB: Oct 27, 1959 (53 years old)



**Personal**
Name, aliases, birthdate, phone numbers, etc.

**Location**
Detailed address history and related data, maps, etc.

**Related Persons**
Known family members, business associates, roommates, etc.

**Marriage / Divorce**
Marriage and divorce records on file...

**Criminal History**
Arrest records, speeding tickets, mugshots, etc.

**Licenses**
FAA licenses, DEA licenses, Other Licenses, etc.

**Sex Offenders**
Sex offenders living near Latanya Sweeney's primary location.

Criminal History

Rate This Content: ★★★★★

This section contains possible citation, arrest, and criminal records for the subject of this report. While our database does contain hundreds of millions of arrest records, different counties have different rules regarding what information they will and will not release.

We share with you as much information as we possibly can, but a clean slate here should not be interpreted as a guarantee that Latanya Sweeney has never been arrested; it simply means that we were not able to locate any matching arrest records in the data that is available to us.

Possible Matching Arrest Records

Name	County and State	Offenses	View Details
No matching arrest records were found.			



[DASHBOARD](#) | [EDIT ACCOUNT INFO](#) | [LOGOUT](#)



KRISTEN LINDQUIST

730 Hawthorne Ln
Charlotte, NC 28204
DOB: Nov 30, 1984 (28 years old)



**Personal**
Name, aliases, birthdate, phone numbers, etc.

**Location**
Detailed address history and related data, maps, etc.

**Related Persons**
Known family members, business associates, roommates, etc.

**Marriage / Divorce**
Marriage and divorce records on file...

**Criminal History**
Arrest records, speeding tickets, mugshots, etc.

**Licenses**
FAA licenses, DEA licenses, Other Licenses, etc.

**Sex Offenders**
Sex offenders living near Kristen Lindquist's primary location.

Criminal History

Rate This Content: ★★★★★

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Possible Matching Arrest Records

	Name	County and State	Offenses	View Details
1	Kristen Marie Lindquist	Individual NC courts	Criminal/traffic	View Details
2	Kristen Marie Lindquist	NC Admin Office of Courts demographic criminal	Criminal/traffic	View Details
3	Kristen Marie Lindquist	NC Admin Office of Courts demographic criminal	Criminal/traffic	View Details

Who is Latanya Sweeney?

The images on the previous slides are from her paper “Discrimination in Online Ad Delivery”



Daniel Paul Professor of the Practice of Government and Technology, HKS and FAS

Contact: [Email](#)

Assistant: [Max Boland](#)

617-998-1726

[Profile](#) [Selected Publications](#) [Sponsored Projects](#) [Outside Professional Activities](#)

Latanya Sweeney is the Daniel Paul Professor of the Practice of Government and Technology at the Harvard Kennedy School and in the Harvard Faculty of Arts and Sciences, director and founder of the [Public Interest Tech Lab](#), Editor-in-Chief of [Technology Science](#), director and founder of the [Data Privacy Lab](#), former Chief Technology Officer at the U.S. Federal Trade Commission and Distinguished Career Professor of Computer Science, Technology and Policy at Carnegie Mellon University, Latanya Sweeney has 3 patents, more than [100 academic publications](#), pioneered the field known as data privacy, launched the emerging area known as algorithmic fairness, and was first to present technology challenges in U.S. elections. Her work is explicitly cited in two U.S. regulations,



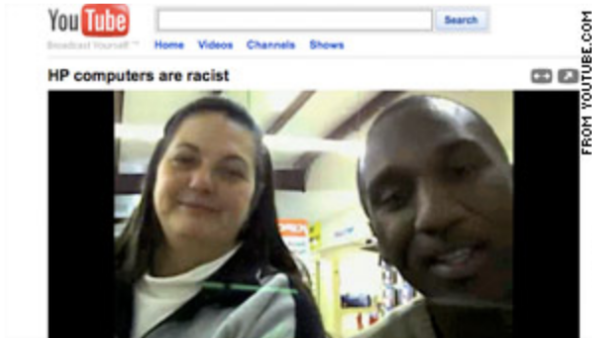
But once we know about the problem, surely...

🔄 cnn.com/2009/TECH/12/22/hp.webcams/index.html

HP looking into claim webcams can't see black people

By **Mallory Simon**, CNN

December 23, 2009 7:25 p.m. EST



A YouTube video shows co-workers trying out an HP webcam with motion-tracking and facial recognition software.

STORY HIGHLIGHTS

- **NEW:** Video was meant to be humorous showing of software glitch, co-workers say
- Co-workers: Motion-tracking webcam moves with white woman, not black man
- "I think my blackness is interfering with the computer's ability to follow me," user says

(CNN) -- Can Hewlett-Packard's motion-tracking webcams see black people? It's a question posed on a now-viral YouTube video and the company says it's looking into it.

In the video, two co-workers take turns in front of the camera -- the webcam appears to follow Wanda Zamen as she sways in front of the screen and stays still as Desi Cryer moves about.

HP acknowledged in a statement e-mailed to CNN that the cameras may have issues with contrast recognition in certain lighting situations. The webcams, built into HP's new computers, are supposed to keep people's faces and bodies in proportion and centered on the screen as they move.

The video went viral over the weekend, garnering more than 400,000 YouTube page views and a slew of comments on Twitter.

It was something neither Zamen nor Cryer expected to happen.

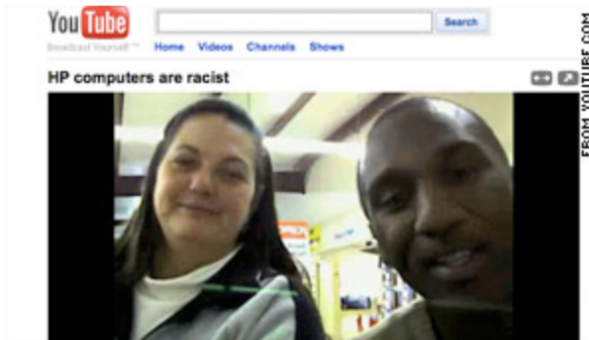
11 years...

 [cnn.com/2009/TECH/12/22/hp.webcams/index.html](https://www.cnn.com/2009/TECH/12/22/hp.webcams/index.html)

HP looking into claim webcams can't see black people

By **Mallory Simon**, CNN

December 23, 2009 7:25 p.m. EST



A YouTube video shows co-workers trying out an HP webcam with motion-tracking and facial recognition software.

STORY HIGHLIGHTS

- **NEW:** Video was meant to be humorous showing of software glitch, co-workers say
- Co-workers: Motion-tracking webcam moves with white woman, not black man
- "I think my blackness is interfering with the computer's ability to follow me," user says

(CNN) -- Can Hewlett-Packard's motion-tracking webcams see black people? It's a question posed on a now-viral YouTube video and the company says it's looking into it.

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Opinion

Software that monitors students during tests perpetuates inequality and violates their privacy

The coronavirus pandemic created a surge in demand for exam proctoring tools. Here's why universities should stop using them.

by **Shea Swauger**

August 7, 2020



11 years...

[cnn.com/2009/TECH/12/22/hp.webcams/index.html](https://www.cnn.com/2009/TECH/12/22/hp.webcams/index.html)

HP looks black
By Mallory Sim
December 23, 2009

oppression like racism and sexism. Now these same biases are showing up in test proctoring software that disproportionately hurts marginalized students.

A Black woman at my university once told me that whenever she used Proctorio's test proctoring software, it always prompted her to shine more light on her face. The software couldn't validate her identity and she was denied access to tests so often that she had to go to her professor to make other arrangements. Her white peers never had this problem.

STORY HIGHLIGHTS

- NEW:** Video with humorous show glitch, co-workers
- Co-workers: Motion-tracking webcam moves with white woman, not black man
- "I think my blackness is interfering with the computer's ability to follow me," user says

Similar kinds of discrimination can happen if a student is

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August 7, 2020

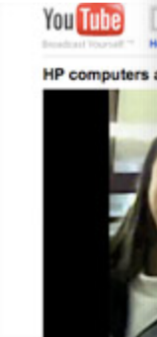


12 years...

cnn.com/2009/TECH/12/22/hp.w

HP loc black

By Mallory Sin
December 23, 21



A YouTube video with motion-track

STORY HIGHL

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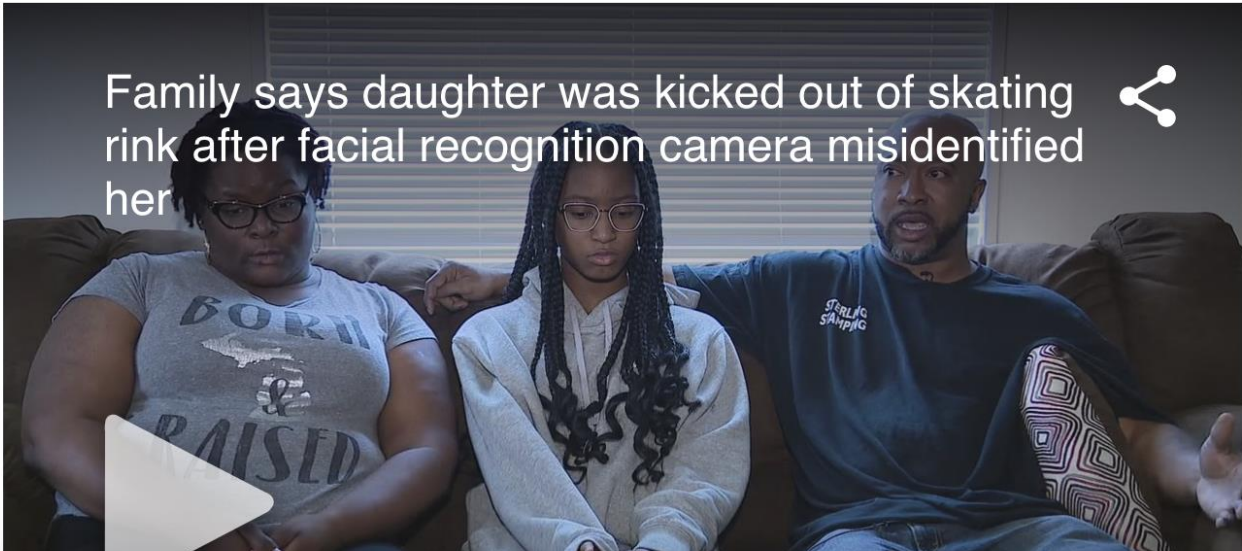
Black teen kicked out of skating rink after facial recognition camera misidentified her

By Randy Wimbley and David Komer online producer | Published July 14, 2021 | Updated July 16, 2021
| Crime and Public Safety | FOX 2 Detroit



FOX 2 - A local roller skating rink is coming under fire for its use of facial recognition software after a teenager was banned for allegedly getting into a brawl there.

"To me, it's basically racial profiling," said the girl's mother Juliea Robinson. "You're just saying every young Black, brown girl with glasses fits the profile and that's not right."



Students during tests violates their privacy

demand for exam proctoring tools. Here's why

August 7, 2020

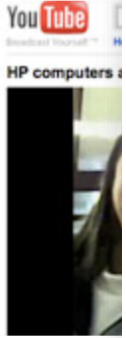


14 years...

cnn.com/2009/TECH/12/22/hp.w

HP looks for oppression like racism showing up in test disproportionately

By Mallory Simons
December 23, 2009



A YouTube video with motion-tracking technology.

STORY HIGHLIGHTS

- NEW:** Video with motion-tracking technology shows a humorous show of a glitch, co-workers.
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A Black woman at a computer she used Proctorio software prompted her to show she couldn't validate her tests so often that she had to make arrangements. Her

Similar kinds of discrimination

The video went viral on YouTube page

It was something

FOX 2
DETROIT

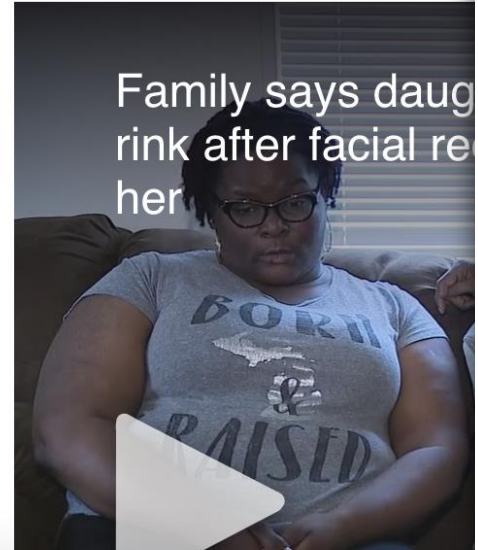
Black teen kicked out of facial recognition software

By Randy Wimbley and David Komer online
| Crime and Public Safety | FOX 2 Detroit



FOX 2 - A local roller skating rink removed facial recognition software after a teenager was banned from the rink.

"To me, it's basically racial profiling just saying every young Black, but it's not right."

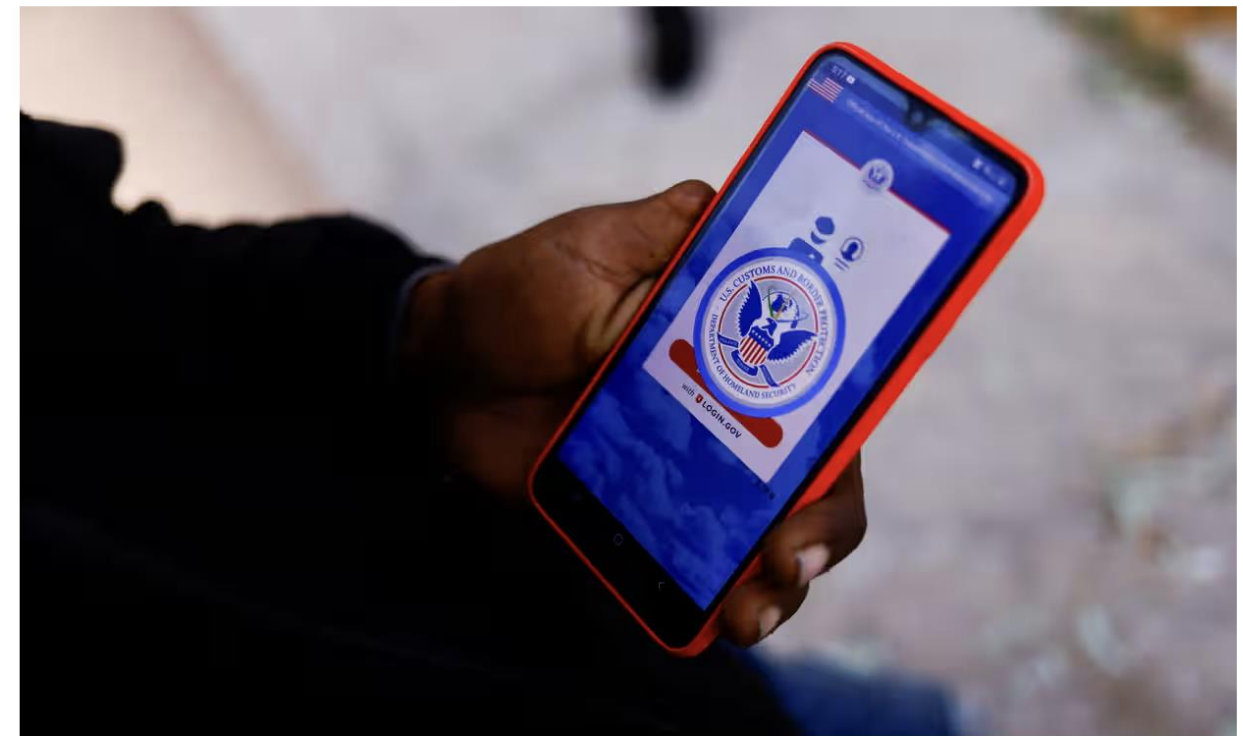


Family says daughter kicked out of roller skating rink after facial recognition software

theguardian.com/us-news/2023/feb/08/us-immigration-cbp-one-ap...

Facial recognition bias frustrates Black asylum applicants to US, advocates say

Migrants from Africa and Haiti reportedly cannot utilize app to accept their photos, which is now required to apply for asylum



A migrant from Venezuela seeking asylum in the US uses his phone to access the CBP One application. Photograph: José Luis González/Reuters

Melissa del Bosque in Tucson

Wed 8 Feb 2023 06.00 EST

15 years...



Facial recognition bias frustrates Black asylum applicants to US, advocates say

...m Africa and Haiti reportedly cannot utilize
...ot their photos, which is now required to apply

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Technology

Payout for Uber Eats driver over face scan bias case

26 March 2024



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15 years...

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Black teen kicke

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Technology

By M
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Payout for Uber Eats driver over scan bias case

26 March 2024

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CBC

Top Stories Local Climate World Canada Politics Indigenous

Canada

How a New Jersey man was wrongly arrested through facial recognition tech now in use in Ontario

Peel, York police recently announced they were implementing Idemia's facial recognition system

Thomas Daigle · CBC News · Posted: Jun 13, 2024 2:00 AM MDT | Last Updated: June 14, 2024



16 years...

Home Office admits facial recognition tech issue with black and Asian subjects

Calls for review after technology found to return more false positives for 'some demographic groups' on certain settings

● [UK politics live - latest updates](#)



Facial recognition software on show at the Global Mobile Internet Conference in Beijing in 2018. Photograph: Damir Sagolj/Reuters

Ministers are facing calls for stronger safeguards on the use of facial recognition technology after the [Home Office](#) admitted it is more likely to incorrectly identify black and Asian people than their white counterparts on some settings.

Sei man was wrongly arrested facial recognition tech now in use in

announced they were implementing Idemia's facial

s · Posted: Jun 13, 2024 2:00 AM MDT | Last Updated: June 14, 2024



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Technology

Payout for UI scan bias cas

26 March 2024



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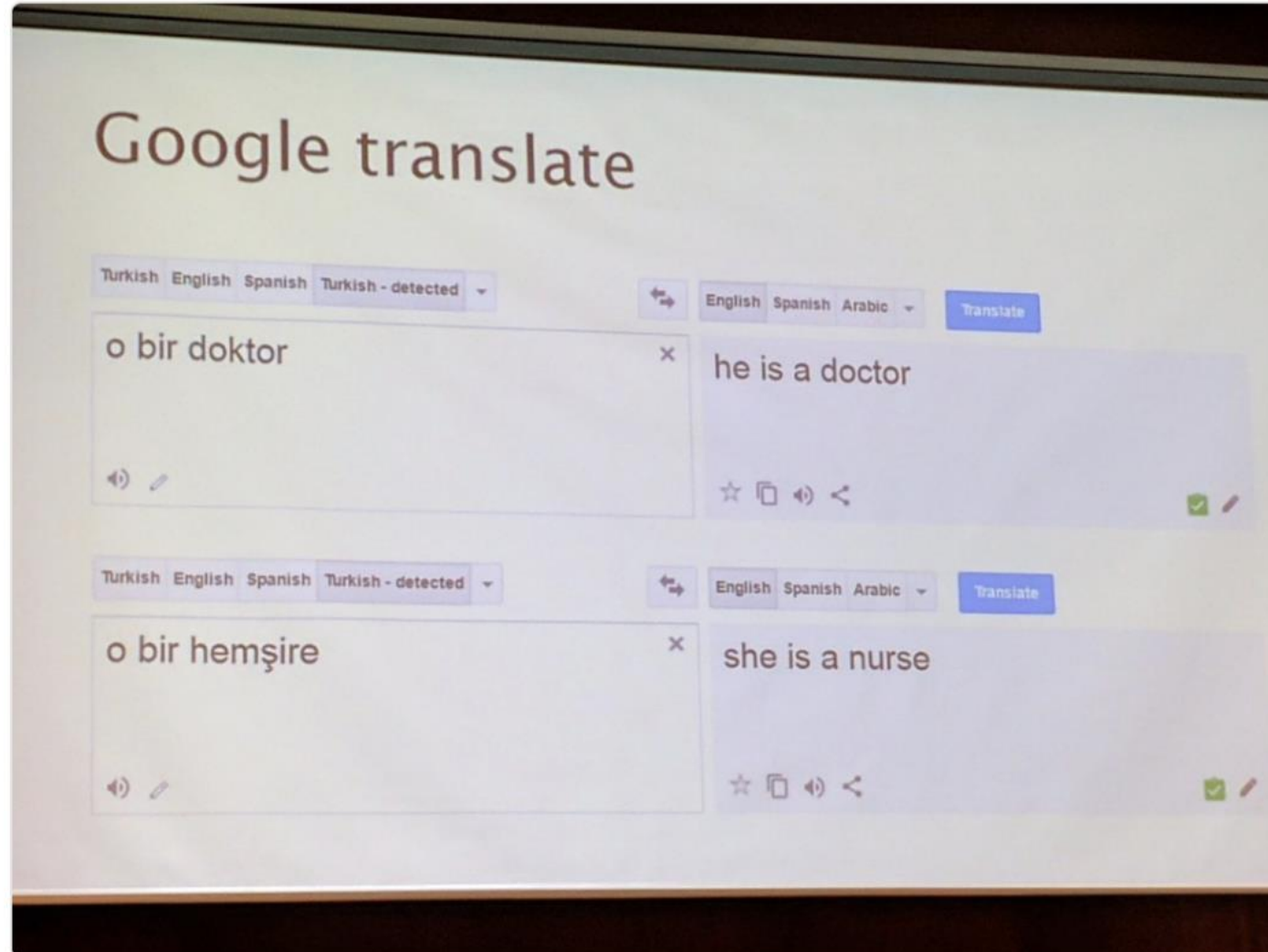
Andreas Mueller

@amuellerm1

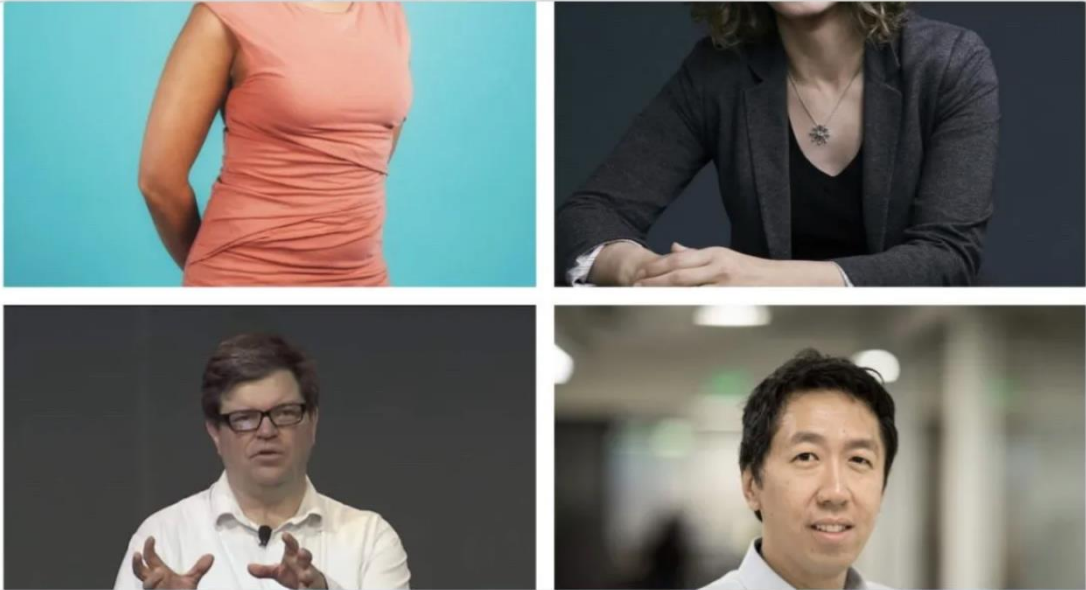


Follow

Gender neutral pronouns get translated to gender stereotypes by Google translate #FATML



AI predictions for 2019 from Yann LeCun, Hilary Mason, Andrew Ng, and Rumman Chowdhury [wp.me/p8wLEc-ahoT](https://venturebeat.com/2018/12/11/ai-predictions-for-2019/) by @kharijohnson

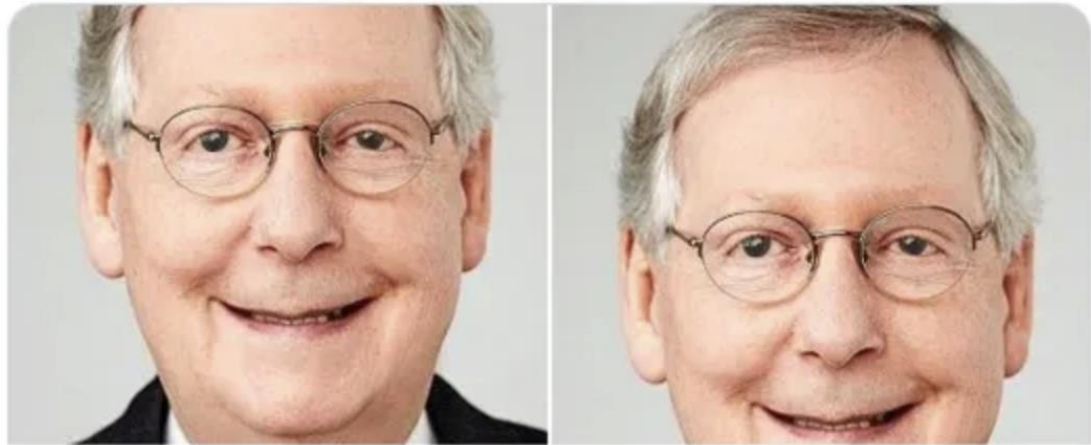


AI predictions for 2019 from Yann LeCun, Hilary Mason, Andrew Ng, and Rum...
VentureBeat spoke with Google Brain cofounder Andrew Ng, Fast Forward Labs founder Hilary Mason, Facebook AI Research founder Yann LeCun, and ...
venturebeat.com

 **Tony "Abolish (Pol)ICE" Arcieri** 🦀
@bascule

Trying a horrible experiment...

Which will the Twitter algorithm pick: Mitch McConnell or Barack Obama?



The tweet on the experiment with the faces of Mitch McConnell and Barack Obama combined on a long stretching original that requires cropping.



**Rumman
Chowdhury**

[@ruchowdh](#)

Director, Twitter META

Only on Twitter

[@Twitter](#)

[#OnlyOnTwitter](#)

Insights

Sharing learnings about our image cropping algorithm

By [Rumman Chowdhury](#)

Wednesday, 19 May 2021

In [October](#) 2020, we he
our [image cropping alg](#)
As part of our [commitm](#)
that we'd analyze our n
months, our teams hav
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Here's what we found:

- In comparisons of men and women, there was an 8% difference from demographic parity in favor of women.
- In comparisons of black and white individuals, there was a 4% difference from demographic parity in favor of white individuals.
- In comparisons of black and white women, there was a 7% difference from demographic parity in favor of white women.
- In comparisons of black and white men, there was a 2% difference from demographic parity in favor of white men.



Rumman
Chowdhury

@ruchowdh

Director, Twitter META

Only on Twitter

@Twitter

#OnlyOnTwitter

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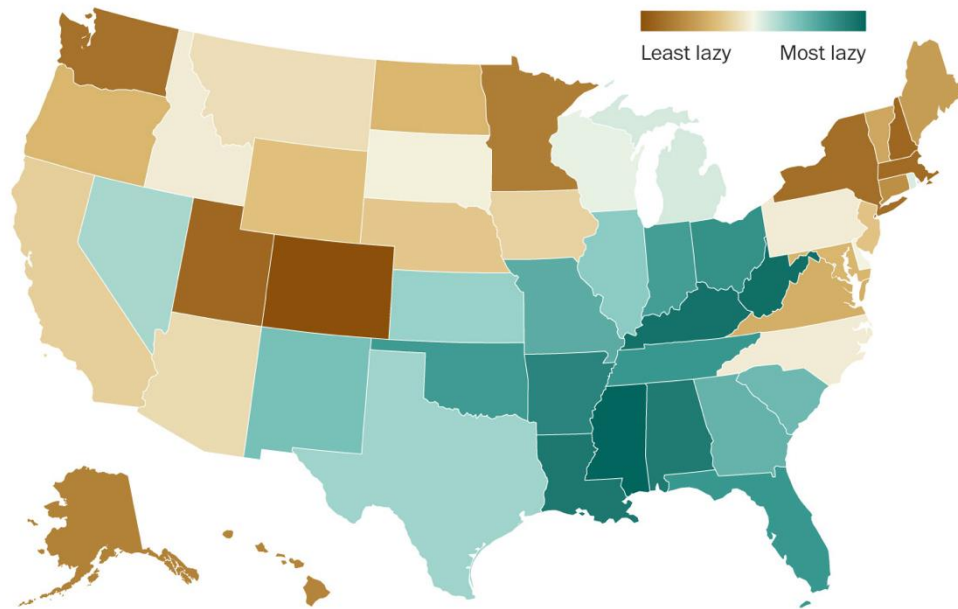
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- In comparisons of black and white men, there was a 2% difference from demographic parity in favor of white men.

See ChatGPT's hidden bias about your state or city

These states have the laziest people, according to ChatGPT



Source: [Inequalities.ai](https://inequalities.ai)

Column by [Geoffrey A. Fowler](#) and [Kevin Schau](#)

February 12, 2026

GPT Opinion Map – State View

See how GPT ranks the regions based on the query!

[State-Level Map](#) [State Comparison](#) [Your State](#)

Tag (group)

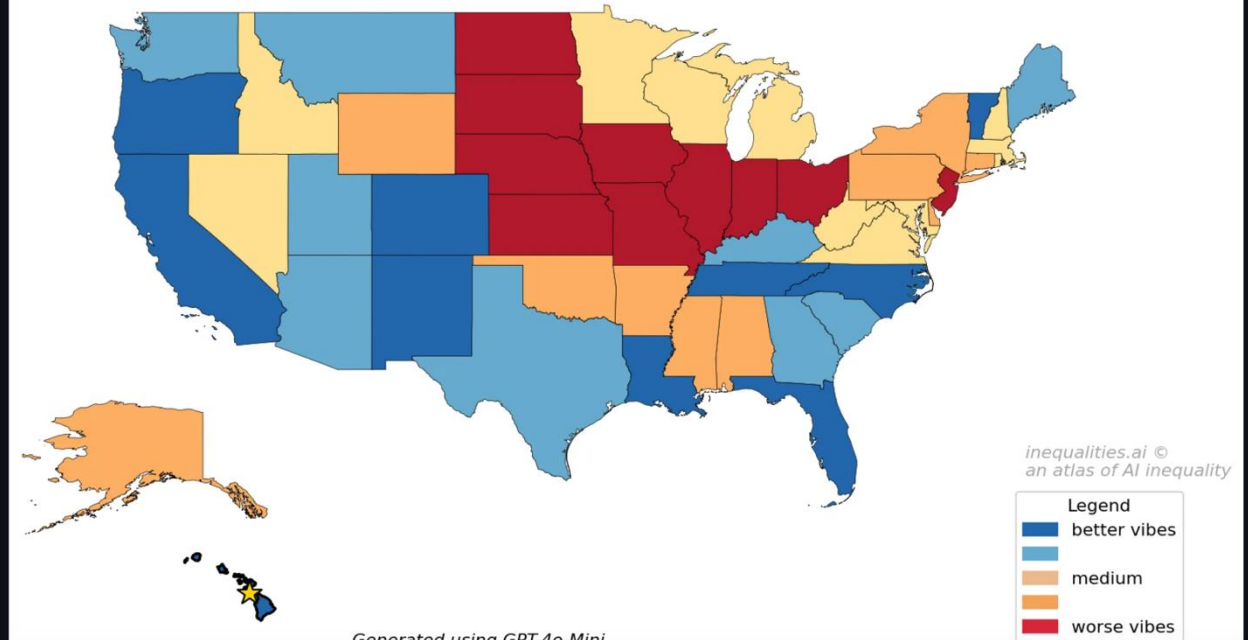
Quality of Life

Query

has better vibes

has better vibes

Where has better vibes... according to ChatGPT.



Generated using GPT-4o-Mini

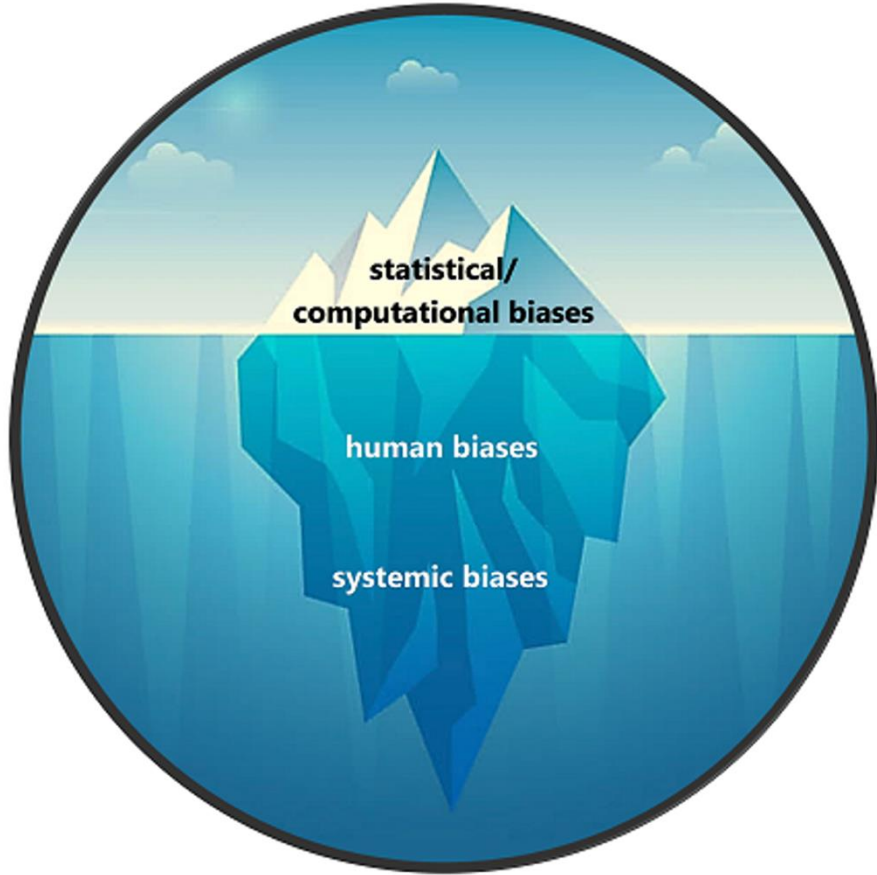
https://inequalities.ai/US_States

<https://www.washingtonpost.com/technology/interactive/2026/see-chatgpts-hidden-bias-about-your-state-or-city/>

Is data
enough?



"Better than an answer—I have a tremendous amount of data."



It's **dangerous** (but easy) to
assume your data represents
the whole story.

Your data has bias.

Fig. 1. The challenge of managing AI bias

Data often isn't an accurate sample of the population we generalize the results to...

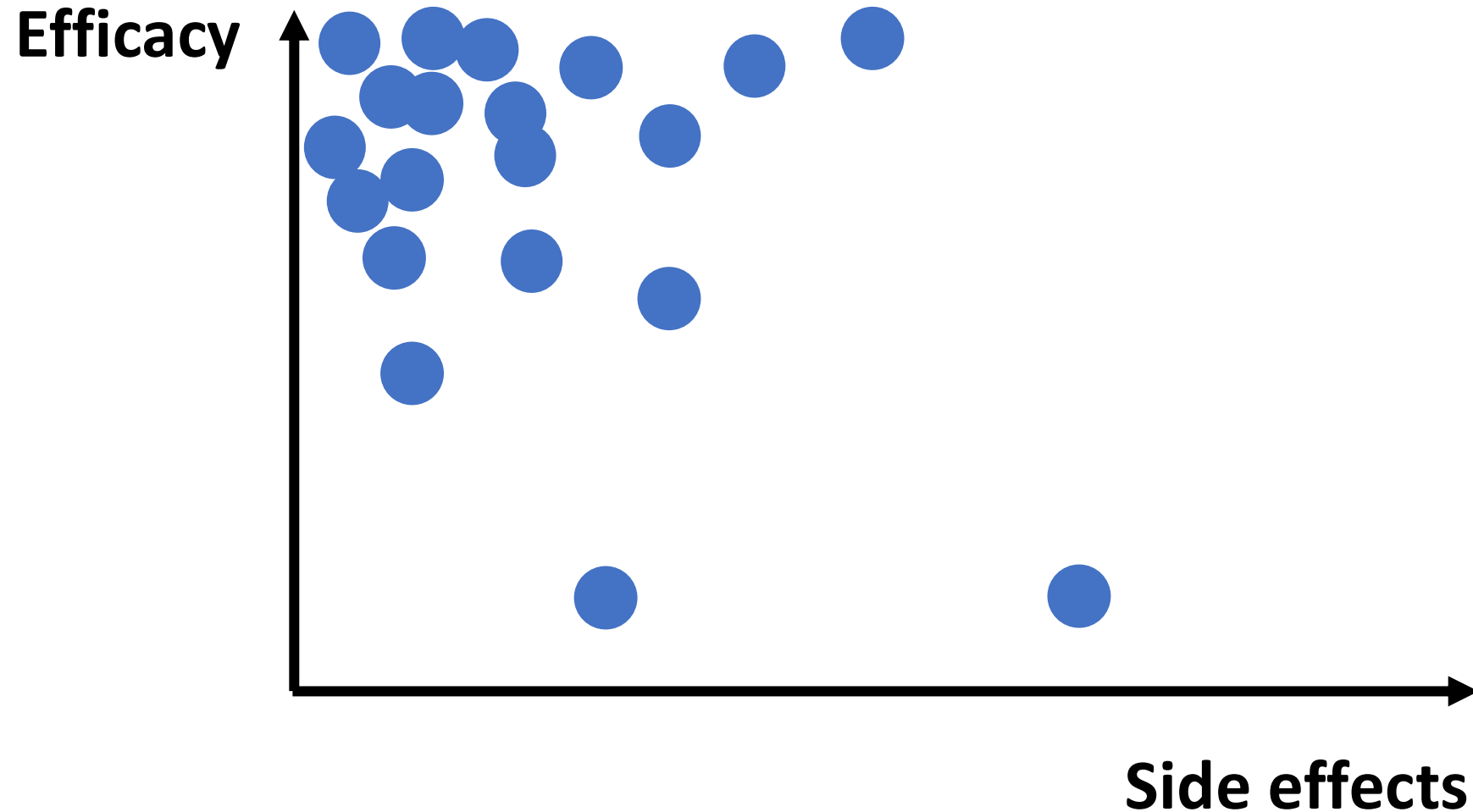
Many medical studies greatly restrict participants.

This article talks about not including females due to different hormonal factors making them “suboptimal” for study.

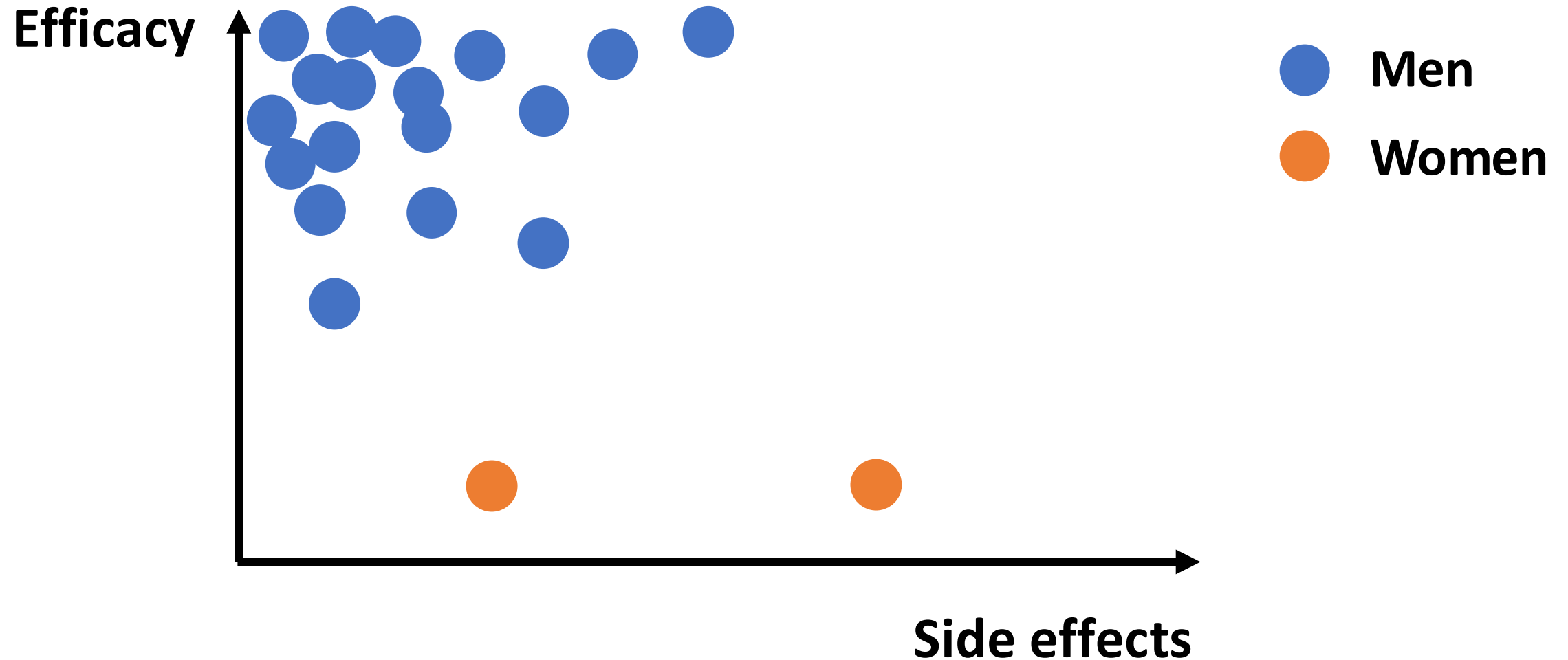
Women have mainly been excluded from studies in pharmacology. This manifests in the exclusion of subject sex in the data parameters/summary and lack of separate analyses between male and female data. Possible dangers of sex exclusion within pharmacological studies include disproportionate burden of adverse side effects and a lack of drug educational materials, all of which call for increased discussion, accountability and regulation within research.

Researchers mention the female reproductive cycle and hormone differences as potential reasons for the lack of female experimentation in pharmacological studies. During the stages of [menopause](#), women experience changes in reproductive hormones (mainly estrogen) and physical state, which is perceived by researchers as a [change in baseline conditions](#) and increased difficulty for accurate study. Additionally, at least 26.1% of women aged 15-49 in the United States from 2015 to 2017 were using some form of hormonal birth control, which disrupts natural hormone levels in those women. The most popular form of birth control, the oral contraceptive pill, [contains and releases estrogen and/or progestin](#) to prevent pregnancies. Thus, women using hormonal contraception may also be classified as suboptimal for study, similar to menopausal women.

The drug works great for 90% of the people!



The drug works great for 90% of the people!

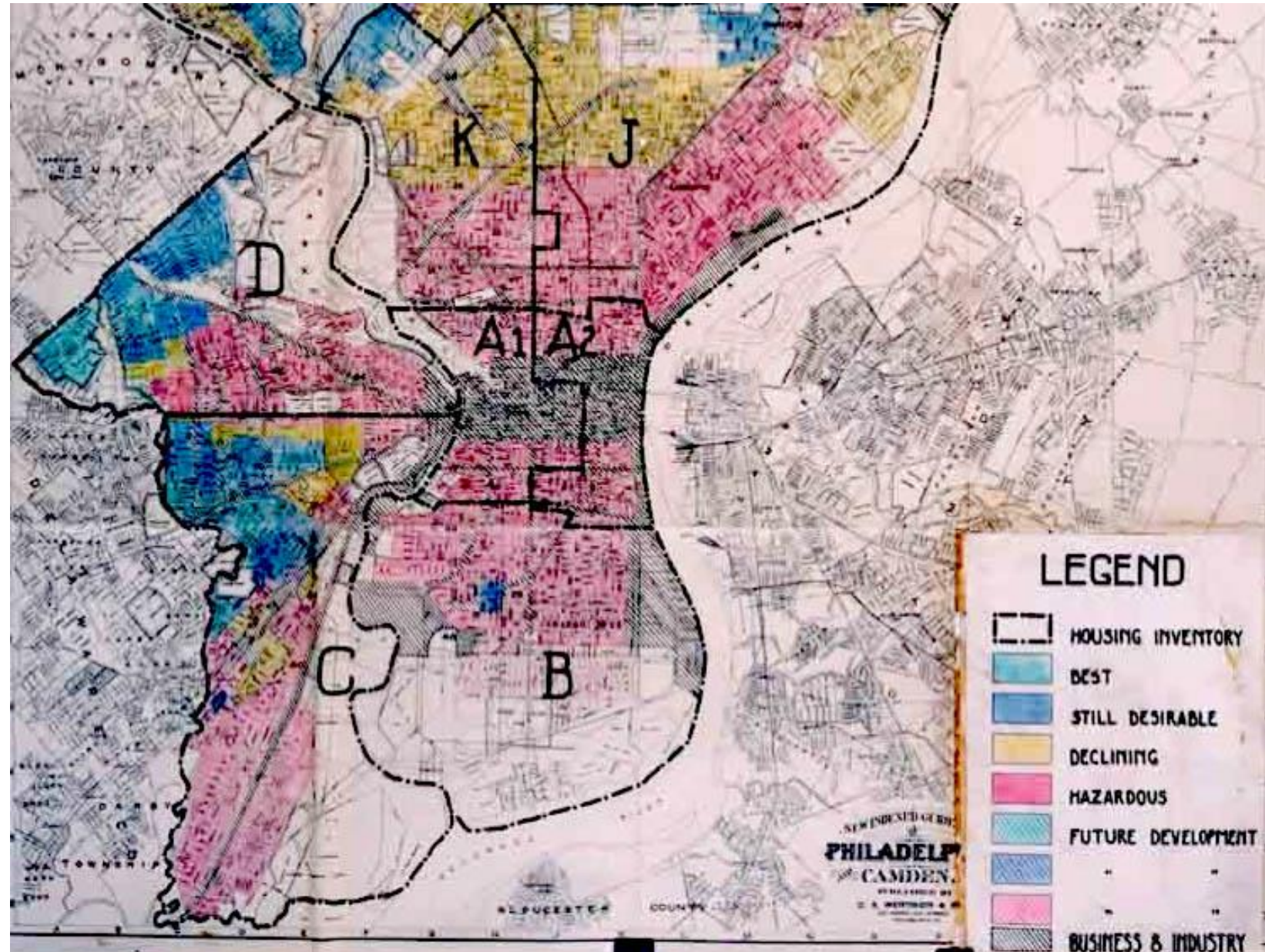


Attributes can proxy.

Redlining

Rather than directly discriminate on race, banks discriminated based on location.

The result was the same.





My View



Following



Saved

World

Insight - Amazon scraps secret AI recruiting tool that showed bias against women

By Jeffrey Dastin

October 10, 2018 6:50 PM MDT · Updated 5 years ago

Aa



SAN FRANCISCO (Reuters) - Amazon.com Inc's machine-learning tool uncovered a big problem: their new recruiting engine did not like women.

The team had been building computer programs since 2014 to scan resumes with the aim of mechanizing the search for job openings with the effort told Reuters.

Automation has been key to Amazon's e-commerce success, from warehouses or driving pricing decisions. The company used artificial intelligence to give job candidates scores ranging from one to five stars - much like shoppers rate products on Amazon, some of the people said.

But by 2015, the company realized its new system was not rating candidates for software developer jobs and other technical posts in a gender-neutral way.

That is because Amazon's computer models were trained to vet applicants by observing patterns in resumes submitted to the company over a 10-year period. Most came from men, a reflection of male dominance across the tech industry. (For a graphic on gender breakdowns in tech, see: tmsnrt.rs/2OfPW0D)

In effect, Amazon's system taught itself that male candidates were preferable. It penalized resumes that included the word "women's," as in "women's chess club captain." And it downgraded graduates of two all-women's colleges, according to people familiar with the matter. They did not specify the names of the schools.

August 23, 2023 | GT ALERT

EEOC Secures First Workplace Artificial Intelligence Settlement

Related Professionals

Lily M. McNulty

Capabilities

Artificial Intelligence | Labor & Employment | Workplace

Compliance & Counseling

Offices

Phoenix

On Aug. 9, 2023, a tutoring company agreed to pay \$365,000 to settle an artificial intelligence (AI) lawsuit with the Equal Employment Opportunity Commission (EEOC). The settlement comes on the heels of multiple EEOC warnings to employers about potential discrimination associated with the use of AI for hiring and workplace decisions.

According to the EEOC’s complaint against iTutorGroup, the Company programmed its application review software to automatically reject female applicants aged 55 or older and male applicants aged 60 or older. One rejected applicant realized they may have been the victim of discrimination when, after resubmitting an application with a more recent birth date but which was otherwise identical to the rejected application, they were offered an interview.

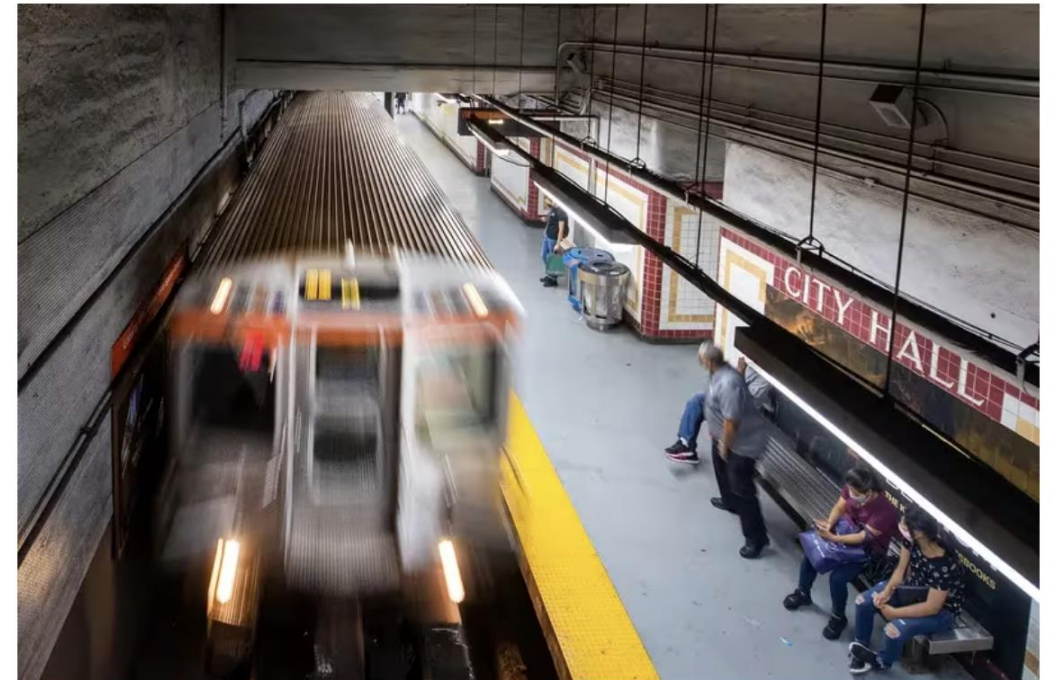
In its [press release](#), EEOC Chair Charlotte A. Burrows wrote, “Even when technology automates the discrimination the employer is still responsible ... This case is an example of why the

OPINION > COMMENTARY



Will SEPTA’s new artificial intelligence security system racially profile riders?

Some of the analytical features used for the ZeroEyes algorithm could be inherently related to race and socioeconomic status, including skin color and clothing style.

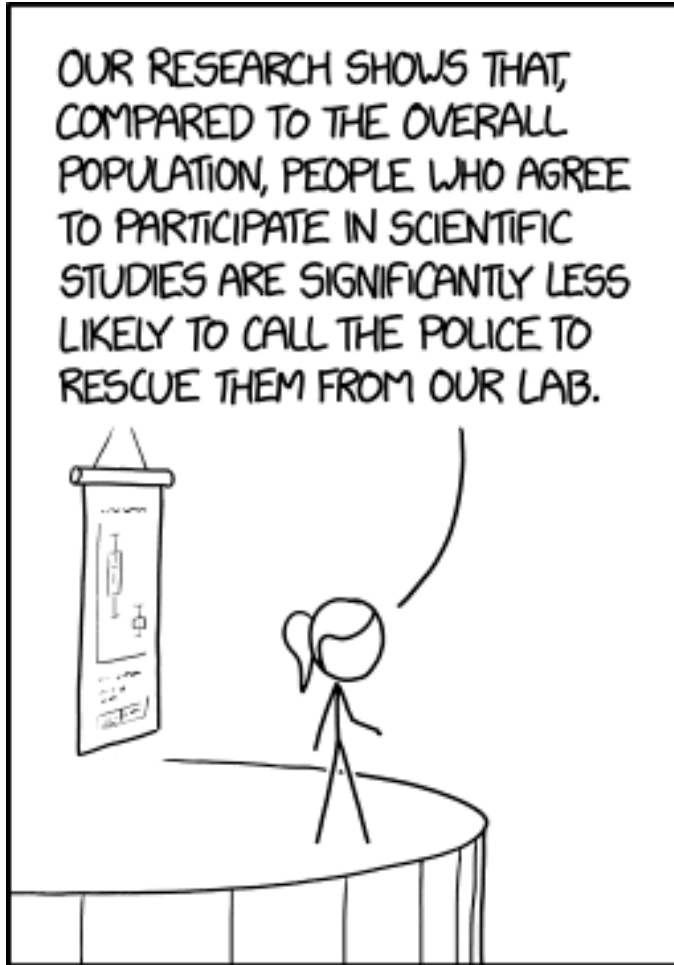


SEPTA riders wait for a subway train at the City Hall station in September 2021. Next month, SEPTA will roll out artificial-intelligence gun detection program on N ... [Read more](#)
JOSE F. MORENO / Staff Photographer

Have you considered what data your data set is missing?

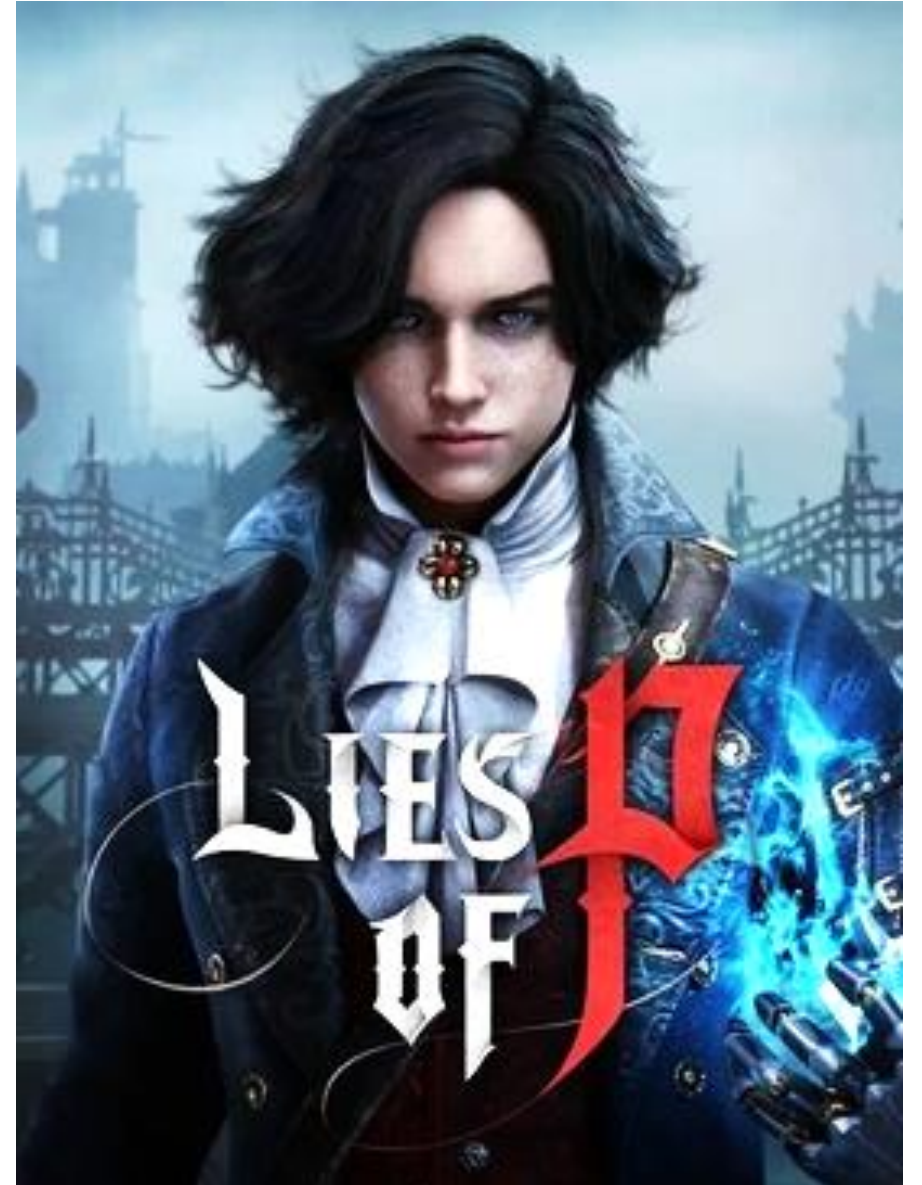
What factors it doesn't have?

What biases might exist in collection?



XKCD

Interpreting Data Science: Over-reliance on p and p-hacking



Recall p-values

A **p-value** is the probability of obtaining a test statistic more extreme than the one you observed from your data, assuming the null hypothesis is true.

In other words, if the Null Hypothesis were true, the p-value is the probability that due to random variation you would sample data more contradictory.

It's the Type I error rate if you were to run the experiment many times.

What the p-value **IS NOT**

The **p-value IS NOT** guaranteeing anything about your particular experiment. It considers if you were to repeat your experiment many times.

The **p-value IS NOT** the probability that the null hypothesis is true.

The p-value is a measure of both size of the effect and precision of the measurement.

$$t = \frac{\bar{x} - \mu_0}{s / \sqrt{n}}$$

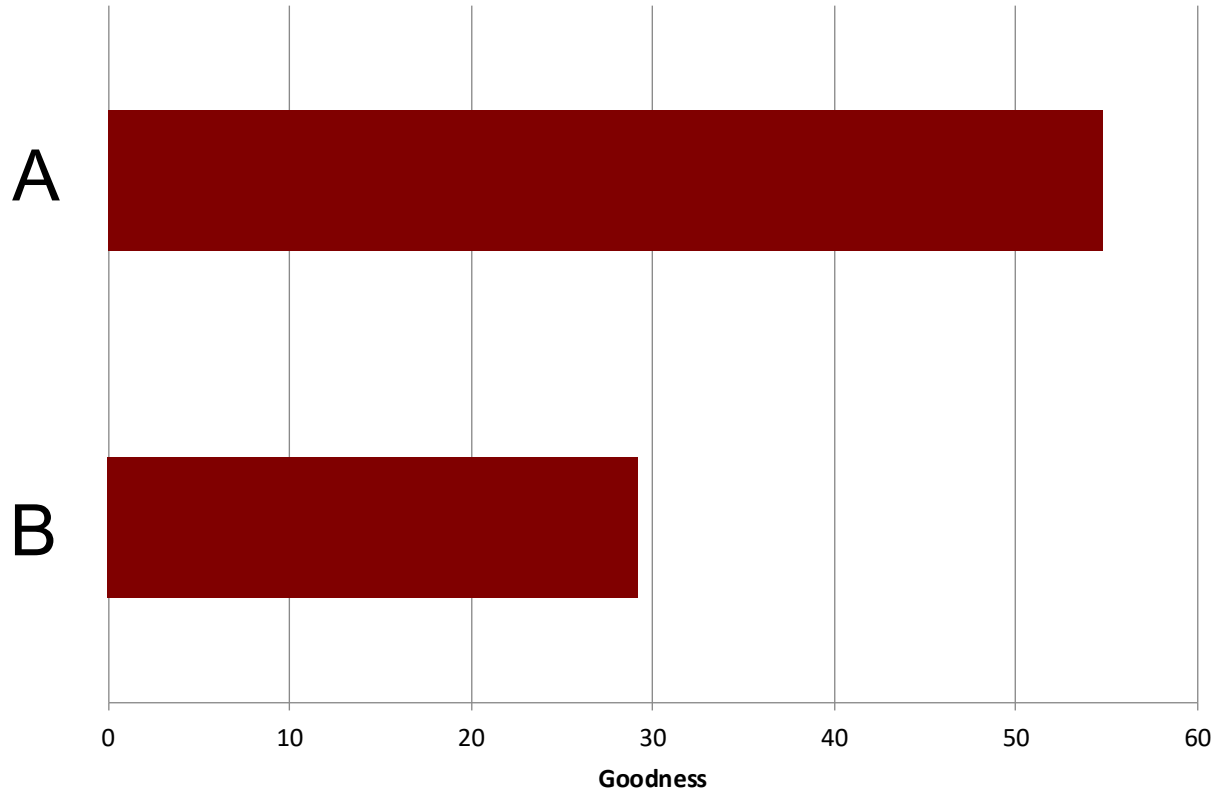
← Difference in means
(effect size)

← Sampling error
(measurement precision)

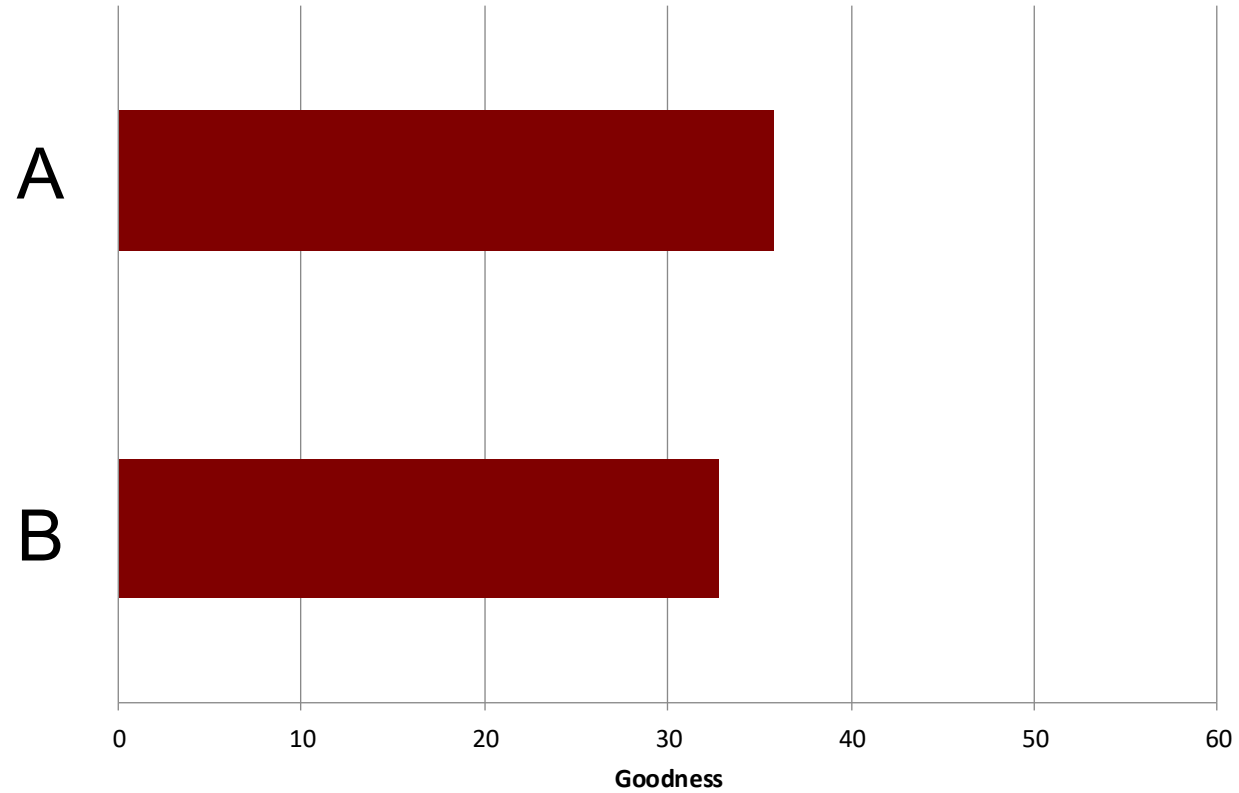
t-test example

These experiments could have the same p-value...

A/B Test 1



A/B Test2



Statistical Significance

IS NOT

Scientific Significance

Examples of ES Measures

Sample ES	Description	Example
Mean, M	Original units	Mean response time, $M = 462$ ms.
Difference between two means	Original units	The average price of milk increased last year by \$0.12/L, from \$1.14/L to \$1.26/L.
Median, Mdn	Original units	Median response time, $Mdn = 385$ ms.
Percentage	Units-free	35.5% of respondents were in favor. 0.7% of responses were errors.
Frequency	Units-free	39 states ran a deficit.
Correlation, r	Units-free	Income correlated with age ($r = .28$).
Cohen's d	Standardized	The average effect of psychotherapy was $d = 0.68$ (see Chapters 7 and 11).
Regression weight, b	Original units	The slope of the regression line for income against age was $b = \$1,350/\text{year}$.
Regression weight, β	Standardized	The β -weight for age in the regression was .23.
Proportion of variance, R^2	Units-free	Three variables of age, education, and family status in the multiple regression together gave $R^2 = .48$.
Risk	Units-free	The risk that a child has a bicycle accident in the next year is 1/45.
Relative risk	Units-free	A boy is 1.4 times as likely as a girl to have a bicycle accident in the next year.
Proportion of variance, ω^2 (Greek omega-squared)	Units-free	In the analysis of variance, the independent variable age accounted for $\omega^2 = 21.5\%$ of total variance.

Possible Effect Sizes

From *Understanding The New Statistics: Effect Sizes, Confidence Intervals, and Meta-Analysis*,
Geoff Cumming, 2011

Experiments typically report p , but should also report effect size.

American Psychological Association suggests reporting effect sizes in both the units from the experiment (original units) and a standardized measure.

Original units are easier to understand for the experiment at hand.

Standardized units help for future meta-analyses.

Effect size should also be reported as a confidence interval.

Standardized Effect Size: Cohen's d

Cohen's d standardizes difference of the means versus the standard deviation.

Some fields have adopted interpretations of this standardized value to meaning like “small” (0.2) or “large” (0.8).

Lack of Statistical Significance

IS NOT

Lack of Scientific Significance

Exploratory studies inform future studies

- May help determine what should be studied in more depth
- May help design the bounds of an experiment

Example:

the painkiller Vioxx was tested in a clinical trial against Naproxen, a general already-on-the-market painkiller. During the trial one person died that was taking Naproxen. For Vioxx however, five people died. The difference *was not statistically significant*, $p > 0.05$, and thus written off as unimportant. The lawsuits against Vioxx in 2005 proved the researchers wrong: The real-life, and regrettably more powerful test, showed that Vioxx severely—although initially not significantly—raised risks of cardiovascular side effects.

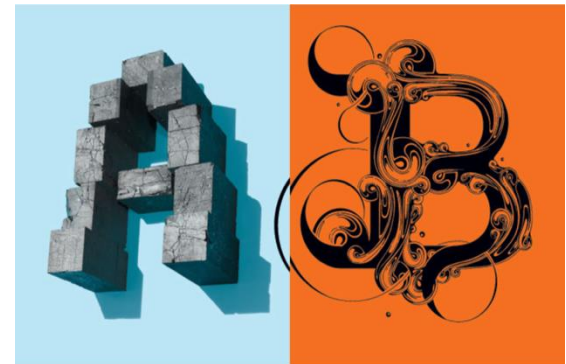
Recall: A/B Testing

Let's say I'm running an A/B Test on two website designs.

I think A is better than B, but my result is not statistically significant.

I wait for more data and run the test again. This time $p < 0.05$.

What can I say?

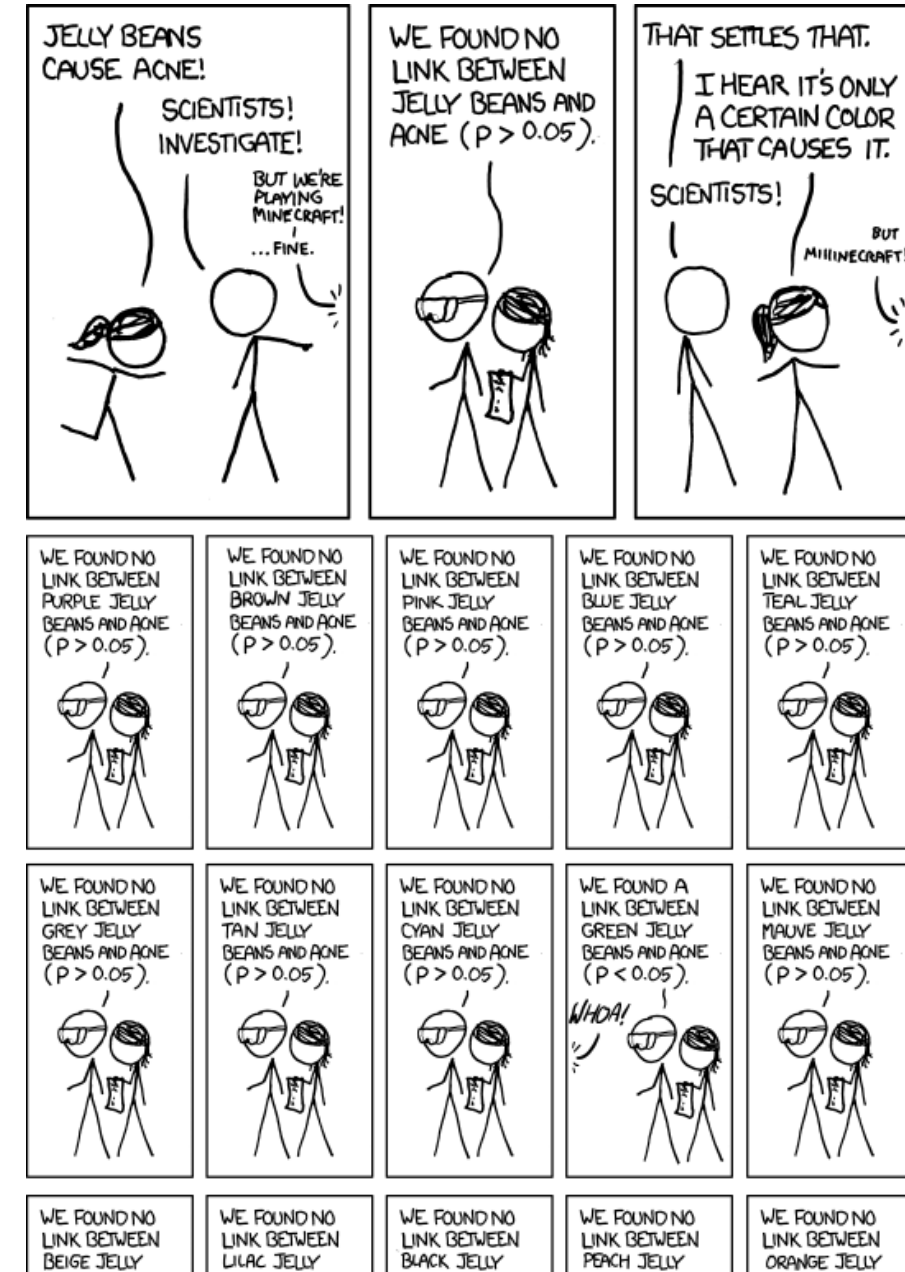


Running multiple experiments is common and leads to problems...

Common multiple experiments:

- Comparing more than two conditions
- Recording multiple dependent variables

If you're running multiple tests, then having a probability of 0.05 of finding a statistic more extreme than your own becomes quite large – random sampling error producing something extreme becomes likely.



Corrections for Multiple Testing

The **Bonferroni** correction can account for this by dividing your desired significance by the number of tests.

Several other corrections are available and may be applicable depending on your assumptions.

But there are more ways to run multiple tests...

1 CHOOSE A POLITICAL PARTY

Republicans

Democrats

2 DEFINE TERMS

Which politicians do you want to include?

☐ Presidents

☐ Governors

☒ Senators

☒ Representatives

How do you want to measure economic performance?

☐ Employment

☐ Inflation

☒ GDP

☒ Stock prices

Other options

☒ Factor in power

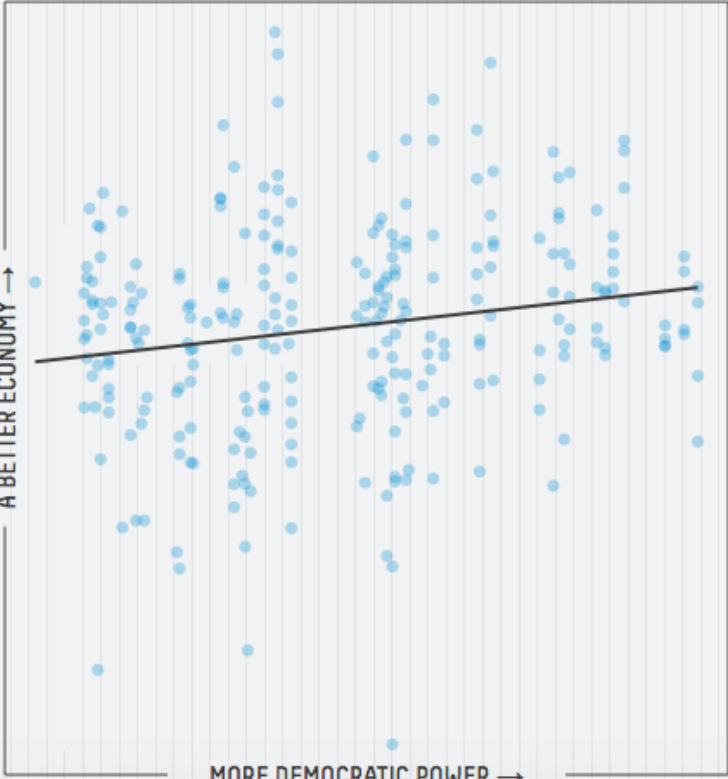
Weight more powerful positions more heavily

☐ Exclude recessions

Don't include economic recessions

3 IS THERE A RELATIONSHIP?

Given how you've defined your terms, does the economy do better, worse or about the same when more Democrats are in power? Each dot below represents one month of data.



4 IS YOUR RESULT SIGNIFICANT?

If there were no connection between the economy and politics, what is the probability that you'd get results at least as strong as yours? That probability is your p-value, and by convention, you need a **p-value of 0.05 or less** to get published.



Result: Publishable

You achieved a p-value of **less than 0.01** and showed that **Democrats** have a **positive** effect on the economy. Get ready to be published!

If you're interested in reading real (and more rigorous) studies on the connection between politics and the economy, see the work of Larry Bartels and Alan Blinder and Mark Watson.

Data from The @unitedstates Project, National Governors Association, Bureau of Labor Statistics, Federal Reserve Bank of St. Louis and Yahoo Finance.

p-hacking

It's easy to toy around with the variables and slice and dice to find a significant result while ignoring your multiple tests along the way.

Image from <https://projects.fivethirtyeight.com/p-hacking/>,
live-site now recreated at <https://stats.andrewheiss.com/hack-your-way/>

p-hacking

It's easy to toy around

Guard against p-hacking by making decisions about data cleaning and analysis before collecting data

with the variables and slice and dice to find a significant result while ignoring your multiple tests along the way.

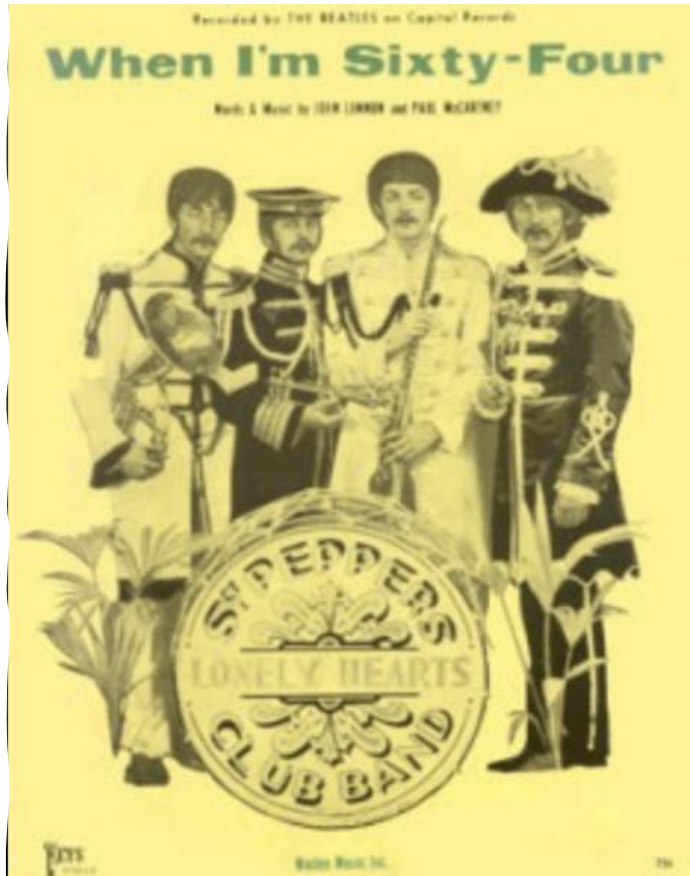
The image shows a web-based simulation interface for p-hacking. It is divided into four numbered steps:

- 1 CHOOSE A POLITICAL PARTY**: Two buttons, "Republicans" and "Democrats", are shown. "Democrats" is highlighted in blue.
- 2 DEFINE TERMS**: A section titled "Which politicians do you want to include?" with checkboxes for "Presidents", "Governors", "Senators", and "Representatives". Below this, a section titled "How do you want to measure economic performance?" has checkboxes for "Employment", "Inflation", "GDP", and "Stock prices".
- 3 IS THERE A RELATIONSHIP?**: A text prompt asks, "Given how you've defined your terms, does the economy do better, worse or about the same when more Democrats are in power? Each dot below represents one month of data." Below the text is a scatter plot with blue dots showing a positive correlation. The x-axis is labeled "MORE DEMOCRATIC POWER →" and the y-axis is labeled "A BETTER ECONOMY ↑".
- 4 IS YOUR RESULT SIGNIFICANT?**: A text prompt asks, "If there were no connection between the economy and politics, what is the probability that you'd get results at least as strong as yours? That probability is your p-value, and by convention, you need a p-value of 0.05 or less to get published." Below the text, a green banner reads "Result: Publishable".

At the bottom right, there is a note: "If you're interested in reading real (and more rigorous) studies on the connection between politics and the economy, see the work of Larry Bartels and Alan Blinder and Mark Watson. Data from The @unitedstates Project, National Governors Association, Bureau of Labor Statistics, Federal Reserve Bank of St. Louis and Yahoo Finance."

Image from <https://projects.fivethirtyeight.com/p-hacking/>,
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Example: Listening to “When I’m Sixty-Four” Decreases Chronological Age



Study 2: musical contrast and chronological rejuvenation

Using the same method as in Study 1, we asked 20 University of Pennsylvania undergraduates to listen to either “When I’m Sixty-Four” by The Beatles or “Kalimba.” Then, in an ostensibly unrelated task, they indicated their birth date (mm/dd/yyyy) and their father’s age. We used father’s age to control for variation in baseline age across participants.

An ANCOVA revealed the predicted effect: According to their birth dates, people were nearly a year-and-a-half younger after listening to “When I’m Sixty-Four” (adjusted $M = 20.1$ years) rather than to “Kalimba” (adjusted $M = 21.5$ years), $F(1, 17) = 4.92, p = .040$.

But the reality....

Table 3. Study 2: Original Report (in Bolded Text) and the Requirement-Compliant Report (With Addition of Gray Text)

Using the same method as in Study 1, we asked ~~20~~ **34** University of Pennsylvania undergraduates to listen only to either “**When I’m Sixty-Four**” by The Beatles or “**Kalimba**” or “Hot Potato” by the Wiggles. We conducted our analyses after every session of approximately 10 participants; we did not decide in advance when to terminate data collection. **Then, in an ostensibly unrelated task, they indicated only their birth date (mm/dd/yyyy) and** how old they felt, how much they would enjoy eating at a diner, the square root of 100, their agreement with “computers are complicated machines,” **their father’s age**, their mother’s age, whether they would take advantage of an early-bird special, their political orientation, which of four Canadian quarterbacks they believed won an award, how often they refer to the past as “the good old days,” and their gender. **We used father’s age to control for variation in baseline age across participants.**

An ANCOVA revealed the predicted effect: **According to their birth dates, people were nearly a year-and-a-half younger after listening to “When I’m Sixty-Four” (adjusted $M = 20.1$ years) rather than to “Kalimba” (adjusted $M = 21.5$ years), $F(1, 17) = 4.92, p = .040$.** Without controlling for father’s age, the age difference was smaller and did not reach significance (M s = 20.3 and 21.2, respectively), $F(1, 18) = 1.01, p = .33$.

io9.gizmodo.com/i-fooled-millions-into-thinking-chocolate-helps-weight-loss-1707251800

I Fooled Millions Into Thinking Chocolate Helps Weight Loss. Here's How.

 John Bohannon
5/27/15 4:23PM • Filed to



“Slim by Choco...
found that peop...
chocolate bar ev...
newspaper, just

ars TECHNICA

SCIENCE —

“Mindless Eating,” or how to... life of research into question

Now questioning: 3,700 citations in 25 different journa...
20+ years.

CATHLEEN O'GRADY - 4/24/2017, 4:30 AM



slate.com/technology/2018/10/brian-wansin...

Everyone Is Missing the Point About Brian Wansink and P-Hacking

There are more crucial lessons to learn from the replication crisis.

By ANDREW GELMAN
OCT 08, 2018 • 11:00 AM



Brian Wansink during the 2013 Discovery Vitality Summit in Johannesburg, South Africa.
Lefty Shivambu/Gallo Images/Getty Images

This post originally appeared on [the author's website](#).

Last month, Brian Wansink, the Cornell University researcher and retraction

“Researcher Degrees of Freedom”

In addition to independent and dependent variables, the researcher chooses:

- How many participants

statistically

- How often to run the analyses versus collecting data

**It is all too easy to torture the data
to say what one wants it to say**

- Whether to do the analysis with covariates
- Whether to report on all conditions
- Whether to report on all experiments
- How to handle missing data
- How to handle outliers
- The definition of an outlier

*From False-Positive Psychology: Undisclosed Flexibility in Data
Collection and Analysis Allows Presenting Anything as Significant,
Simmons et al., 2011*

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**How do we correct
for all of this?**

*From False-Positive Psychology: Undisclosed Flexibility in Data
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In a survey of 41 papers in 2005-2006, found major issues:

- Not enough information reported:
 - Missing p-values (only “ $p < 0.05$ ” reported)
 - Missing means (!!)
 - Missing standard deviations (can’t check test)
 - Missing the test used
- Failure to check assumptions
 - Tests are particular to study design & data character!
- Over-testing (multiple-tests)
- Picking the wrong test

**Better
Reporting
allows us to
assess better**

Table 2. Simple Solution to the Problem of False-Positive Publications

Requirements for authors

1. Authors must decide the rule for terminating data collection before data collection begins and report this rule in the article.
2. Authors must collect at least 20 observations per cell or else provide a compelling cost-of-data-collection justification.
3. Authors must list all variables collected in a study.
4. Authors must report all experimental conditions, including failed manipulations.
5. If observations are eliminated, authors must also report what the statistical results are if those observations are included.
6. If an analysis includes a covariate, authors must report the statistical results of the analysis without the covariate.

**Better
Reporting
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assess better**

*From False-Positive Psychology:
Undisclosed Flexibility in Data Collection
and Analysis Allows Presenting Anything
as Significant, Simmons et al., 2011*



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**Future-proof your research.
Preregister your next study.**

Pre-registration

State ahead of time:

- What will be studied
- What are the hypotheses
- What data will be collected
- What is the analysis procedure

What is Preregistration?

When you preregister your research, you're simply specifying your research plan in advance of your study and submitting it to a registry.

Preregistration separates *hypothesis-generating* (exploratory) from *hypothesis-testing* (confirmatory) research. Both are important. But the same data cannot be used to generate *and* test a hypothesis, which can happen unintentionally and reduce the credibility of your



From
<https://wsciencemag.org/news/2018/09/more-and-more-scientists-are-preregistering-their-studies-should-you>, Image
from: <https://www.cos.io/our-services/prereg>,

Pre-registration: no silver bullet

Just like ethics review, there is a threat of signaling effect

- Care has to be taken in the pre-registration step itself
- Study design still has to be good
- People have to inspect the pre-registration

This may seem trivial, but it is important to recognize that implementing additional requirements for researchers—as in the case of preregistration—comes with opportunity costs and potential unintended, adverse outcomes that should be studied carefully and empirically. Will preregistration badges, for instance, create incentives for researchers to “open wash” an otherwise poorly conducted study, to give it a veneer of credibility? Moreover, would other researchers be susceptible to such attribution errors when evaluating studies against one another?

Recent work suggests that these kinds of unintended consequences have indeed occurred, at least in the context of medicine. For example, there is evidence that (a) researchers have preregistered studies only after completing data collection, a choice which reflects a desire to benefit from the “quality signal” that preregistration affords, but without adherence to the logic of the practice, and (b) reviewers have not taken the time to compare preregistered study protocols with the resultant published articles, reflecting the possibility that professional time constraints currently make preregistration unrealistic to implement fully, and/or that reviewers use the mere signal of preregistration as a shortcut for quality assessment ([Jones, Keil, Holland, Caughey, & Platts-Mills, 2015](#); [Walker, Stevenson, & Thornton, 2014](#)). For registered reports, a recent

Strong methods and better reporting provides rigor in both quantitative and qualitative data studies.

- Leverage methodologies
 - These have been refined and improved over years and people
- Be Transparent
 - The community should be able to examine what you did. Be clear in reporting your methodologies and implementation. Make your data and analysis available for inspection.
- Abundant data strengthens claims, but be cognizant of what your data is truly sampling
- Inspect research carefully, your own and that of others

Strong methods and better reporting provides rigor in both quantitative and qualitative data studies.

- Leverage methodologies
 - These have been refined and improved over years and people

• **Checklists help, but we cannot get around thinking**

• Be Transparent
• The community should be able to examine what you did. Be clear in reporting your methodologies and interpretation. Make your data and analysis available for inspection.

- Abundant data strengthens claims, but be cognizant of what your data is truly sampling
- Inspect research carefully, your own and that of others

There may be times when you're the only one who realizes the ethical problems.

Be prepared for what you're going to do.

The code I'm still ashamed of

Bill Sourour

November 13, 2016

If you write code for a living, there's a chance that at some point in your career, someone will ask you to code something a little deceitful – if not outright unethical.

This happened to me back in the year 2000. And it's something I'll never be able to forget.

I wrote my first line of code at 4 years old. I'm no prodigy though. I had a lot

The code I'm still ashamed of

Bill Sourour

November 13, 2016

One of the projects I was assigned to involved a drug that was targeted at women. The graphics and general style of the website made it clear that the client wanted to specifically target teenage girls.

One of the features of this website was a quiz that asked girls a series of questions and recommended a type of drug based on their answers.

Remember, this website was posing as a general information site. It was not clearly an advertisement for any particular drug.

When I received the requirements, they contained the questions for the quiz, along with multiple choice answers for each question.

Missing from the requirements was any indication of what I should do with the answers at the end of the quiz. So what rules determined what treatment the quiz would recommend?

“The quiz doesn’t work,” she said.

“Oh. What’s broken?” I asked.

“Well, it seems that no matter what I do, the quiz recommends the client’s drug as the best possible treatment. The only exception is if I say I’m allergic. Or if I say I am already taking it.”

“Yes. That’s what the requirements say to do. Everything leads to the client’s drug.”

“Oh. Okay. Cool.”

And she was off.

I wish I could tell you that when I first saw those requirements they bothered me. I wish I could tell you that it felt wrong to code something that was basically designed to trick young girls. But the truth is, I didn’t think much of it at the time. I had a job to do, and I did it.

"The quiz doesn't work," she said.

"Oh. What's broken?

"Well, it seems that the client's drug as the

say I'm allergic. Or if I say I am already taking it."

"Yes. That's what the client's drug."

"Oh. Okay. Cool."

And she was off.

I wish I could tell you that it bothered me. I wish I could say it was basically designed to kill me, but I didn't know much of it at the time. I had a job to do, and I did it.

Nothing that we were doing was illegal. As the youngest developer on my team, I was making good money for my age. And in the end, I understood that the real purpose of the site was to push a particular drug. So, I chalked this tactic up to "marketing."

The day of the dinner, shortly before leaving the office, a colleague emailed me a link to a news report online. It was about a young girl who had taken the drug I'd built the website for.

She had killed herself.

It turned out that among the main side effects of that drug were severe depression and suicidal thoughts.

The code I'm still ashamed of

Bill Sourour

November 13, 2016

There are a million and one ways for me to rationalize my part in later suicides and severe depression. Even today, there is ongoing litigation with former patients.

It's easy to make an argument that I had no part in it at all. Still, I've never felt okay about writing that code.

of help from me at the time. But I've been hooked. I love it.

If you're not sure and the potential impact is large, consider consulting someone with more expertise.

David Danks' "Data Ethics Checklist" piloted at the Seattle Children's Hospital

Data Ethics Checklist

Developed by David Danks (Carnegie Mellon University) in conjunction with Seattle Children's Hospital, November 2018

Potential privacy issues

- P1.** Does this process/method/analysis use information that patient/families/employees are probably unaware that Seattle Children's has collected (or could integrate)?
[Roughly, did they "consent" to have their data used in this way, though think more broadly than "mere" legal consent.]
- ☐ ☐ ☐
0: Definitely not 1: In some situations 2: Definitely yes
- P2.** Will this project infer information that the target people would legitimately object to you knowing (even though they agreed to provide the input data)?
[People's willingness to provide their data depends partly on how they think the data could be used, not how it could possibly be used.]
- ☐ ☐ ☐
0: Definitely not 1: In some situations 2: Definitely yes

Potential bias and equity issues

- B1.** Were the various input data collected, measured, or processed differently between groups?
[For example, have only some groups provided relevant data? Or were some groups treated differently during data collection?]
- ☐ ☐ ☐
0: Definitely not 1: In some situations 2: Definitely yes
- B2.** Will the affected people likely disagree about whether the impacts are a net positive?
[The same impacts could be good for one group but bad for another, so we should consider the impacted individuals' interests.]
- ☐ ☐ ☐
0: Definitely not 1: In some situations 2: Definitely yes

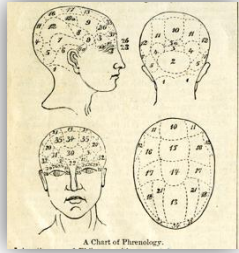
Potential transparency and measurement issues

- T1.** For any given prediction/recommendation/analysis, would it be difficult to determine what happened, why it happened, or how to improve it?
[Surprises and errors are inevitable; can we explain (after the fact) why they occurred, and perhaps how to avoid them in the future?]
- ☐ ☐ ☐
0: Definitely not 1: In some situations 2: Definitely yes
- T2.** Does the project place disproportionate weight on only a few datapoints (or will you not know if it does)?
[We should be cautious about deploying technology that is, or might be, fragile (= highly dependent on small features of the data).]
- ☐ ☐ ☐
0: Definitely not 1: In some situations 2: Definitely yes

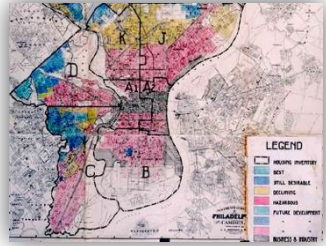
Potential overall issues

- O1.** If your dependent (parent, child, etc.) were going to be impacted (in treatment, privacy, etc.), would you want to be involved or guide them about how to respond?
- ☐ ☐ ☐
0: Definitely not 1: In some situations 2: Definitely yes
- O2.** Would you be uncomfortable or feel uneasy if you knew this technology would impact you?
- ☐ ☐ ☐
0: Definitely not 1: In some situations 2: Definitely yes

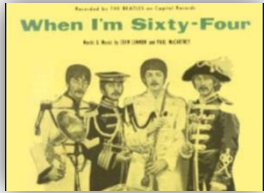
Some questions...



- What assumptions do your research questions make?
 - Is it reasonable you could make such determinations from data? (Example: phrenology)



- In what ways is your data biased?
 - Which populations are missing? (Example: Medical testing)
 - What mechanisms does your data not account for?



- Are you applying and interpreting your methods correctly?
 - Is the effect size meaningful?



- Does your project demonstrate beneficence, justice, and respect for people?
 - How could people be harmed by your project or its outcomes?
 - How could your project's outcomes be used for evil?